

# LLaMA: Open and Efficient Foundation Language Models

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## Abstract

We introduce LLaMA, a collection of foundation language models ranging from 7B to 65B parameters. We train our models on trillions of tokens, and show that it is possible to train state-of-the-art models using publicly available datasets exclusively, without resorting to proprietary and inaccessible datasets. In particular, LLaMA-13B outperforms GPT-3 (175B) on most benchmarks, and LLaMA-65B is competitive with the best models, Chinchilla-70B and PaLM-540B. We release all our models to the research community<sup>1</sup>.

## 1 Introduction

Large Languages Models (LLMs) trained on massive corpora of texts have shown their ability to perform new tasks from textual instructions or from a few examples (Brown et al., 2020). These few-shot properties first appeared when scaling models to a sufficient size (Kaplan et al., 2020), resulting in a line of work that focuses on further scaling these models (Chowdhery et al., 2022; Rae et al., 2021). These efforts are based on the assumption that more parameters will lead to better performance. However, recent work from Hoffmann et al. (2022) shows that, for a given compute budget, the best performances are not achieved by the largest models, but by smaller models trained on more data.

The objective of the scaling laws from Hoffmann et al. (2022) is to determine how to best scale the dataset and model sizes for a particular *training* compute budget. However, this objective disregards the *inference* budget, which becomes critical when serving a language model at scale. In this context, given a target level of performance, the preferred model is not the fastest to train but the fastest at inference, and although it may be cheaper to train a large model to reach a certain level of

performance, a smaller one trained longer will ultimately be cheaper at inference. For instance, although Hoffmann et al. (2022) recommends training a 10B model on 200B tokens, we find that the performance of a 7B model continues to improve even after 1T tokens.

The focus of this work is to train a series of language models that achieve the best possible performance at various inference budgets, by training on more tokens than what is typically used. The resulting models, called *LLaMA*, ranges from 7B to 65B parameters with competitive performance compared to the best existing LLMs. For instance, LLaMA-13B outperforms GPT-3 on most benchmarks, despite being 10× smaller. We believe that this model will help democratize the access and study of LLMs, since it can be run on a single GPU. At the higher-end of the scale, our 65B-parameter model is also competitive with the best large language models such as Chinchilla or PaLM-540B.

Unlike Chinchilla, PaLM, or GPT-3, we only use publicly available data, making our work compatible with open-sourcing, while most existing models rely on data which is either not publicly available or undocumented (e.g. “Books – 2TB” or “Social media conversations”). There exist some exceptions, notably OPT (Zhang et al., 2022), GPT-NeoX (Black et al., 2022), BLOOM (Scao et al., 2022) and GLM (Zeng et al., 2022), but none that are competitive with PaLM-62B or Chinchilla.

In the rest of this paper, we present an overview of the modifications we made to the transformer architecture (Vaswani et al., 2017), as well as our training method. We then report the performance of our models and compare with others LLMs on a set of standard benchmarks. Finally, we expose some of the biases and toxicity encoded in our models, using some of the most recent benchmarks from the responsible AI community.

# LLaMA：开放且高效的基础语言模型

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## 摘要

我们推出了LLaMA，这是一组参数规模从70亿参数模型到650亿参数模型不等的基础语言模型。我们在数万亿个标记上训练这些模型，并证明了仅依靠公开可用的数据集即可训练出处于当前最优水平的语言模型，无需依赖那些专有且无法获取的数据集。具体而言，LLaMA-13B在大多数基准测试中的表现均优于GPT-3（1750亿参数模型），而LLaMA-65B则能与最优秀的模型Chinchilla-70B以及PaLM-540B相抗衡。我们将所有模型都开源给科研界使用<sup>1</sup>。

## 1 引言

在海量文本语料库上训练得到的大语言模型（大语言模型）已展现出能够根据文本指令或少量示例来完成新任务的能力（Brown等人，2020年）。这种少样本学习任务的能力最初是在将模型规模扩大到足够大的时候出现的（Kaplan等人，2020年），由此催生了一系列致力于进一步扩展这类模型规模的研究工作（Chowdhery等人，2022年；Rae等人，2021年）。这些研究都基于这样一个假设：更多的参数能带来更好的性能。然而，Hoffmann等人（2022年）的最新研究表明，在给定的计算资源限制下，最佳性能并非来自规模最大的模型，而是来自在更多数据上训练得到的较小模型。

Hoffmann等人（2022年）提出的缩放定律旨在确定如何针对特定的训练计算预算，最优地调整数据集规模与模型规模。然而，这一目标并未考虑推理预算问题，而在大规模部署语言模型时，后者显得尤为关键。在此背景下，若以特定的性能水平为目标，更优的模型并非训练速度最快的那个，而是推理速度最快的那个；尽管为了达到某一性能水平而训练大型模型可能会成本更低，

性能而言，即便训练时间更长，规模较小的模型在推理阶段的成本最终也会更低。例如，Hoffmann等人（2022年）建议在2000亿个标记上训练100亿参数模型，但我们发现，即使处理1万亿个标记，70亿参数模型的性能依然能够持续提升。

本研究的重点在于训练一系列语言模型，通过使用远超常规用量的标记进行训练，从而在不同的推理预算下实现最佳性能。这些被命名为*LLaMA*的模型参数量介于70亿参数模型到650亿参数之间，其性能已能与当前最优秀的 大语言模型相媲美。例如，尽管LLaMA-13B模型的规模比GPT-3 10×更小，但在大多数测试基准上的表现却更佳。我们认为，由于该模型仅需在单块GPU上即可运行，因此有望推动更多人能够使用和研究 大语言模型。在更高参数量的层级上，我们开发的650亿参数模型同样能与Chinchilla或PaLM-540B等顶尖的大语言模型一较高下。

与Chinchilla、PaLM或GPT-3不同，我们的模型仅使用公开可获取的数据，这一做法符合开源理念，而现有的大多数模型所依赖的数据要么无法公开获取，要么缺乏相关文档记载（例如‘书籍——2TB’或‘社交媒体对话’）。虽然也存在一些例外，比如OPT（Zhang等人，2022年）、GPT-NeoX（Black等人，2022年）、BLOOM（Scao等人，2022年）以及GLM（Zeng等人，2022年），但这些模型都无法与PaLM-62B或Chinchilla相抗衡。

在本文的其余部分，我们将概述对变换器架构所做的修改（Vaswani等人，2017年），同时介绍我们的训练方法。随后，我们会汇报这些模型的性能，并在一系列标准基准测试中将其与其他大语言模型进行对比。最后，借助责任AI领域最新推出的部分基准测试，我们会揭示这些模型中存在的偏见与毒性问题。

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<sup>1</sup>https://github.com/facebookresearch/llama

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https://github.com/facebookresearch/llama



## 2 Approach

Our training approach is similar to the methods described in previous work (Brown et al., 2020; Chowdhery et al., 2022), and is inspired by the Chinchilla scaling laws (Hoffmann et al., 2022). We train large transformers on a large quantity of textual data using a standard optimizer.

### 2.1 Pre-training Data

Our training dataset is a mixture of several sources, reported in Table 1, that cover a diverse set of domains. For the most part, we reuse data sources that have been leveraged to train other LLMs, with the restriction of only using data that is publicly available, and compatible with open sourcing. This leads to the following mixture of data and the percentage they represent in the training set:

**English CommonCrawl [67%].** We preprocess five CommonCrawl dumps, ranging from 2017 to 2020, with the CCNet pipeline (Wenzek et al., 2020). This process deduplicates the data at the line level, performs language identification with a fastText linear classifier to remove non-English pages and filters low quality content with an n-gram language model. In addition, we trained a linear model to classify pages used as references in Wikipedia v.s. randomly sampled pages, and discarded pages not classified as references.

**C4 [15%].** During exploratory experiments, we observed that using diverse pre-processed CommonCrawl datasets improves performance. We thus included the publicly available C4 dataset (Raffel et al., 2020) in our data. The preprocessing of C4 also contains deduplication and language identification steps: the main difference with CCNet is the quality filtering, which mostly relies on heuristics such as presence of punctuation marks or the number of words and sentences in a webpage.

**Github [4.5%].** We use the public GitHub dataset available on Google BigQuery. We only kept projects that are distributed under the Apache, BSD and MIT licenses. Additionally, we filtered low quality files with heuristics based on the line length or proportion of alphanumeric characters, and removed boilerplate, such as headers, with regular expressions. Finally, we deduplicate the resulting dataset at the file level, with exact matches.

**Wikipedia [4.5%].** We add Wikipedia dumps from the June-August 2022 period, covering 20

| Dataset       | Sampling prop. | Epochs | Disk size |
|---------------|----------------|--------|-----------|
| CommonCrawl   | 67.0%          | 1.10   | 3.3 TB    |
| C4            | 15.0%          | 1.06   | 783 GB    |
| Github        | 4.5%           | 0.64   | 328 GB    |
| Wikipedia     | 4.5%           | 2.45   | 83 GB     |
| Books         | 4.5%           | 2.23   | 85 GB     |
| ArXiv         | 2.5%           | 1.06   | 92 GB     |
| StackExchange | 2.0%           | 1.03   | 78 GB     |

Table 1: **Pre-training data.** Data mixtures used for pre-training, for each subset we list the sampling proportion, number of epochs performed on the subset when training on 1.4T tokens, and disk size. The pre-training runs on 1T tokens have the same sampling proportion.

languages, which use either the Latin or Cyrillic scripts: bg, ca, cs, da, de, en, es, fr, hr, hu, it, nl, pl, pt, ro, ru, sl, sr, sv, uk. We process the data to remove hyperlinks, comments and other formatting boilerplate.

**Gutenberg and Books3 [4.5%].** We include two book corpora in our training dataset: the Gutenberg Project, which contains books that are in the public domain, and the Books3 section of ThePile (Gao et al., 2020), a publicly available dataset for training large language models. We perform deduplication at the book level, removing books with more than 90% content overlap.

**ArXiv [2.5%].** We process arXiv Latex files to add scientific data to our dataset. Following Lewkowycz et al. (2022), we removed everything before the first section, as well as the bibliography. We also removed the comments from the .tex files, and inline-expanded definitions and macros written by users to increase consistency across papers.

**Stack Exchange [2%].** We include a dump of Stack Exchange, a website of high quality questions and answers that covers a diverse set of domains, ranging from computer science to chemistry. We kept the data from the 28 largest websites, removed the HTML tags from text and sorted the answers by score (from highest to lowest).

**Tokenizer.** We tokenize the data with the byte-pair encoding (BPE) algorithm (Sennrich et al., 2015), using the implementation from Sentence-Piece (Kudo and Richardson, 2018). Notably, we split all numbers into individual digits, and fallback to bytes to decompose unknown UTF-8 characters.

## 2 方法

我们的训练方案与以往研究中所描述的方法类似（Brown等人，2020年；Chowdhery等人，2022年），其设计灵感源自Chinchilla模型的扩展规律（Hoffmann等人，2022年）。该方法会使用标准的优化器，在海量文本数据上训练出规模更大的变换器架构。

### 2.1 预训练数据

我们的训练数据集由多种来源的数据混合而成，相关情况详见表 1，这些数据覆盖了多种不同的领域。在大多数情况下，我们会复用那些已被用于训练其他大语言模型的数据源，但仅使用公开可用且符合开源要求的数据。由此形成了以下的数据组合以及它们在训练集中所占的比例：

**英文版CommonCrawl数据集 [67%].** 我们采用了CCNet流程（Wenzek等人，2020年）对2017年至2020年间的五批CommonCrawl数据dump进行预处理。该流程会在行级去除重复数据，利用fastText线性分类器进行语言识别以剔除非英语页面，并通过n-gram语言模型过滤掉质量较低的内容。此外，我们还训练了一个线性模型，用于区分维基百科中作为参考的页面与随机抽取的页面，同时将未被识别为参考的页面直接丢弃。

**C4 [15%].** 在探索性实验中，我们发现使用多种经过预处理的CommonCrawl数据集能够提升模型性能，因此我们在数据集中加入了公开可用的C4数据集（Raffel等人，2020年）。C4的预处理过程还包括去重以及语言识别步骤；其与CCNet的主要区别在于质量过滤机制，后者主要依赖标点符号的存在情况以及网页中的单词和句子数量等启发式规则来进行筛选。

**Github [4.5%].** 我们使用了存储在Google BigQuery上的公共GitHub数据集。仅保留了那些采用Apache、BSD和MIT许可证发布的项目。此外，我们还根据行长度或字母数字字符的比例运用启发式规则筛选掉了质量较低的文件，并通过正则表达式去除了页头等冗余内容。最后，在文件层面对处理后的数据集进行完全匹配的去重操作。

**维基百科 [40.5%].** 我们加入了2022年6月至8月期间收录的维基百科数据，涵盖了20种使用拉丁字母或西里尔字母的语言。

| 数据集            | 采样比例、训练轮数、磁盘容量 |      |        |
|----------------|----------------|------|--------|
| CommonCrawl数据集 | 67.0%          | 1.10 | 3.3 TB |
| C4             | 15.0%          | 1.06 | 783 GB |
| Github         | 4.5%           | 0.64 | 328 GB |
| 维基百科           | 4.5%           | 2.45 | 83 GB  |
| 书籍             | 4.5%           | 2.23 | 85 GB  |
| ArXiv          | 2.5%           | 1.06 | 92 GB  |
| StackExchange  | 2.0%           | 1.03 | 78 GB  |

表1: **预训练数据。**用于预训练的数据混合集；针对每个子集，我们列出了其采样比例、在1.4万亿个标记的数据量上对该子集进行的训练轮数，以及对应的磁盘存储大小。在1万亿个标记数据量上进行的预训练也采用相同的采样比例。

这些语言包括保加利亚语、加泰罗尼亚语、捷克语、丹麦语、德语、英语、西班牙语、法语、克罗地亚语、匈牙利语、意大利语、荷兰语、波兰语、葡萄牙语、罗马尼亚语、俄语、斯洛文尼亚语、塞尔维亚语、瑞典语以及乌克兰语。在处理这些数据时，我们会去除其中的超链接、评论以及其他格式化模板内容。

**古腾堡计划与Books3数据集3 [40.5%].** 我们在训练数据集中包含了两个书籍语料库：一个是包含公共领域书籍的古腾堡计划，另一个则是用于训练大型语言模型的公开数据集ThePile中的Books3部分（高等人，2020年）。我们会对书籍层面进行去重处理，剔除内容重叠度超过90%的书籍。

**ArXiv [20.5%].** 我们会处理ArXiv LaTeX文件，以便将其中的科学数据添加到我们的数据集中。依据Lewkowycz等人（2022年）的研究方法，我们删除了每份文件第一节之前的所有内容以及参考文献列表。同时，我们也去除了.tex文件中的评论，以及用户为提升论文一致性而编写的内联展开定义和宏指令。

**Stack Exchange [2%].** 我们在此提供了Stack Exchange的完整数据集，该网站汇聚了大量高质量的问题与答案，涵盖从计算机科学到化学等众多不同领域。这些数据选自规模最大的28个网站，其中的HTML标签已被移除，答案则按照得分从高到低进行了排序。

**分词器。**我们采用字节对编码（BPE）算法（Sennrich等人，2015年）对训练数据进行分词处理，所使用的实现来自Sentence-Piece算法（Kudo和Richardson，2018年）。需要特别说明的是，我们会将所有数字拆分为单独的数字位，而对于无法识别的UTF-8字符，则转而使用字节来进行分解。

| params | dimension | $n$ heads | $n$ layers | learning rate | batch size | $n$ tokens |
|--------|-----------|-----------|------------|---------------|------------|------------|
| 6.7B   | 4096      | 32        | 32         | $3.0e^{-4}$   | 4M         | 1.0T       |
| 13.0B  | 5120      | 40        | 40         | $3.0e^{-4}$   | 4M         | 1.0T       |
| 32.5B  | 6656      | 52        | 60         | $1.5e^{-4}$   | 4M         | 1.4T       |
| 65.2B  | 8192      | 64        | 80         | $1.5e^{-4}$   | 4M         | 1.4T       |

Table 2: **Model sizes, architectures, and optimization hyper-parameters.**

Overall, our entire training dataset contains roughly 1.4T tokens after tokenization. For most of our training data, each token is used only once during training, with the exception of the Wikipedia and Books domains, over which we perform approximately two epochs.

## 2.2 Architecture

Following recent work on large language models, our network is based on the transformer architecture (Vaswani et al., 2017). We leverage various improvements that were subsequently proposed, and used in different models such as PaLM. Here are the main difference with the original architecture, and where we were found the inspiration for this change (in bracket):

**Pre-normalization [GPT3].** To improve the training stability, we normalize the input of each transformer sub-layer, instead of normalizing the output. We use the RMSNorm normalizing function, introduced by Zhang and Sennrich (2019).

**SwiGLU activation function [PaLM].** We replace the ReLU non-linearity by the SwiGLU activation function, introduced by Shazeer (2020) to improve the performance. We use a dimension of  $\frac{2}{3}4d$  instead of  $4d$  as in PaLM.

**Rotary Embeddings [GPTNeo].** We remove the absolute positional embeddings, and instead, add rotary positional embeddings (RoPE), introduced by Su et al. (2021), at each layer of the network.

The details of the hyper-parameters for our different models are given in Table 2.

## 2.3 Optimizer

Our models are trained using the AdamW optimizer (Loshchilov and Hutter, 2017), with the following hyper-parameters:  $\beta_1 = 0.9, \beta_2 = 0.95$ . We use a cosine learning rate schedule, such that the final learning rate is equal to 10% of the maximal learning rate. We use a weight decay of 0.1 and gradient clipping of 1.0. We use 2,000 warmup

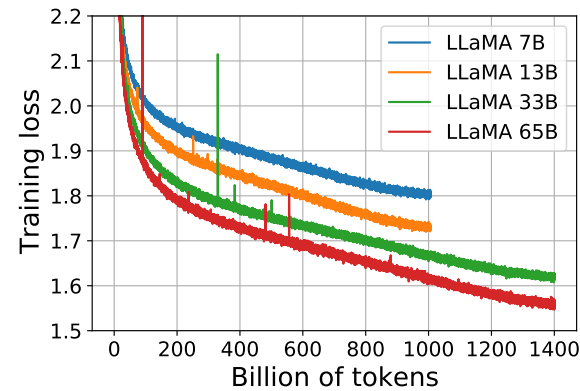


Figure 1: **Training loss over train tokens for the 7B, 13B, 33B, and 65B models.** LLaMA-33B and LLaMA-65B were trained on 1.4T tokens. The smaller models were trained on 1.0T tokens. All models are trained with a batch size of 4M tokens.

steps, and vary the learning rate and batch size with the size of the model (see Table 2 for details).

## 2.4 Efficient implementation

We make several optimizations to improve the training speed of our models. First, we use an efficient implementation of the causal multi-head attention to reduce memory usage and runtime. This implementation, available in the xformers library,<sup>2</sup> is inspired by Rabe and Staats (2021) and uses the backward from Dao et al. (2022). This is achieved by not storing the attention weights and not computing the key/query scores that are masked due to the causal nature of the language modeling task.

To further improve training efficiency, we reduced the amount of activations that are recomputed during the backward pass with checkpointing. More precisely, we save the activations that are expensive to compute, such as the outputs of linear layers. This is achieved by manually implementing the backward function for the transformer layers, instead of relying on the PyTorch autograd. To fully benefit from this optimization, we need to

<sup>2</sup><https://github.com/facebookresearch/xformers>

| 参数数量 | 维度   | $n$ heads | $n$ layers | 学习率         | 批量大小 | $n$ 标记 |
|------|------|-----------|------------|-------------|------|--------|
| 6.7B | 4096 | 32        | 32         | $3.0e^{-4}$ | 4M   | 1.0T   |
| 130亿 | 5120 | 40        | 40         | $3.0e^{-4}$ | 4M   | 1.0T   |
| 325亿 | 6656 | 52        | 60         | $1.5e^{-4}$ | 4M   | 1.4T   |
| 652亿 | 8192 | 64        | 80         | $1.5e^{-4}$ | 4M   | 1.4T   |

表2: 模型规模、架构以及优化超参数。

总体而言, 经过分词处理后, 我们的整个训练数据集大约包含1.4万亿个标记。在大部分训练数据中, 每个标记在训练过程中仅被使用一次, 不过维基百科和书籍类数据除外, 针对这两类数据我们大约会进行两轮训练轮数。

## 2.2 架构

基于近期在大型语言模型领域的研究成果, 我们的网络采用了transformer架构 (Vaswani等人, 2017年)。我们借鉴了随后被提出并应用于PaLM等不同模型中的各类改进方案。以下是该架构与原始架构的主要差异, 以及我们从中获得灵感的具体来源 (标注在括号内):

**预归一化 [GPT3].** 为提升训练稳定性, 我们会对每个transformer子层的输入进行归一化处理, 而非对输出进行归一化。此处所采用的归一化函数为RMSNorm归一化函数, 这一函数由张与Sennrich(2019年)提出。

**SwiGLU激活函数 [PaLM].** 为进一步提升性能, 我们用SwiGLU激活函数替换了原有的ReLU非线性函数, 该激活函数由Shazeer (2020年)提出。此外, 我们采用的维度为 $\frac{2}{3}4d$ , 而非PaLM中所使用的 $4d$ 。

**旋转位置嵌入 [GPTNeo].** 我们去掉了原有的绝对位置嵌入, 转而在网络的每一层都引入了旋转位置嵌入 (即RoPE), 这一技术由苏等人(2021年)提出。

我们各类语言模型的超参数详情详见表 2。

## 2.3 优化器

我们的模型是使用AdamW OPT模型 (Loshchilov与Hutter) 进行训练的, 相关研究发表于2017年, 所采用的超参数如下:  $\beta_1 = 0.9, \beta_2 = 0.95$ 。我们采用了余弦学习率调度机制, 使得最终的学习率等于最大学习率的10%。同时设置权重衰减值为0, 梯度裁剪值为1, 以及0。此外, 我们还使用了2,000的预热策略。

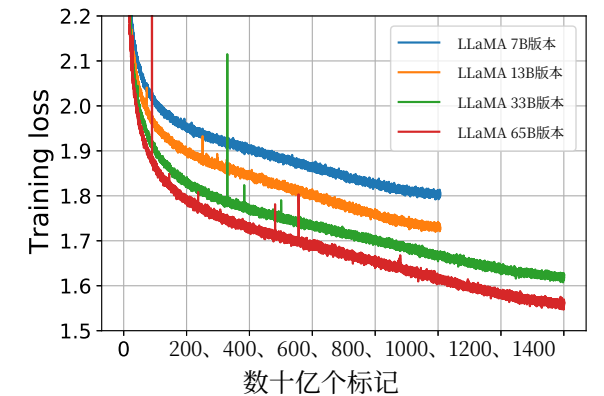


图1: 70亿参数模型、130亿参数模型、330亿参数模型以及650亿参数模型的训练损失随训练标记数量的变化情况。LLaMA-33B版本和LLaMA-65B版本是在1.4万亿个标记上训练的, 而较小规模的模型则在1万亿个标记上训练。所有模型均采用400万个标记作为批量大小进行训练。

训练步数会随模型规模的不同而变化, 学习率与批量大小也会依据模型大小进行调整 (详情请参见表2)。

## 2.4 高效实现方式

为提升模型的训练速度, 我们进行了多项优化。首先, 我们采用了高效的因果多头注意力实现方案, 以此降低内存占用和运行时间。这种实现方式存在于xformers库中,<sup>2</sup> 其设计灵感来源于Rabe与Staats (2021年), 同时借鉴了Dao等人 (2022年) 的研究成果。该方案的实现原理在于不存储注意力权重, 也不计算因语言建模任务的因果特性而被屏蔽的关键值/查询值分数。

为进一步提升训练效率, 我们通过检查点技术减少了在反向传播过程中需要重新计算的激活值数量。更具体而言, 我们会保存那些计算成本较高的激活值, 比如线性层的输出。为此, 我们手动实现了Transformer层的反向传播函数, 而非依赖PyTorch的自动求导功能。要充分受益于这一优化措施, 我们还需要……

<sup>2</sup><https://github.com/facebookresearch/xformers>



|            |      | BoolQ       | PIQA        | SIQA        | HellaSwag   | WinoGrande  | ARC-e       | ARC-c       | OBQA        |
|------------|------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| GPT-3      | 175B | 60.5        | 81.0        | -           | 78.9        | 70.2        | 68.8        | 51.4        | 57.6        |
| Gopher     | 280B | 79.3        | 81.8        | 50.6        | 79.2        | 70.1        | -           | -           | -           |
| Chinchilla | 70B  | 83.7        | 81.8        | 51.3        | 80.8        | 74.9        | -           | -           | -           |
| PaLM       | 62B  | 84.8        | 80.5        | -           | 79.7        | 77.0        | 75.2        | 52.5        | 50.4        |
| PaLM-cont  | 62B  | 83.9        | 81.4        | -           | 80.6        | 77.0        | -           | -           | -           |
| PaLM       | 540B | <b>88.0</b> | 82.3        | -           | 83.4        | <b>81.1</b> | 76.6        | 53.0        | 53.4        |
| LLaMA      | 7B   | 76.5        | 79.8        | 48.9        | 76.1        | 70.1        | 72.8        | 47.6        | 57.2        |
|            | 13B  | 78.1        | 80.1        | 50.4        | 79.2        | 73.0        | 74.8        | 52.7        | 56.4        |
|            | 33B  | 83.1        | 82.3        | 50.4        | 82.8        | 76.0        | <b>80.0</b> | <b>57.8</b> | 58.6        |
|            | 65B  | 85.3        | <b>82.8</b> | <b>52.3</b> | <b>84.2</b> | 77.0        | 78.9        | 56.0        | <b>60.2</b> |

Table 3: **Zero-shot performance on Common Sense Reasoning tasks.**

reduce the memory usage of the model by using model and sequence parallelism, as described by [Korthikanti et al. \(2022\)](#). Moreover, we also overlap the computation of activations and the communication between GPUs over the network (due to all\_reduce operations) as much as possible.

When training a 65B-parameter model, our code processes around 380 tokens/sec/GPU on 2048 A100 GPU with 80GB of RAM. This means that training over our dataset containing 1.4T tokens takes approximately 21 days.

### 3 Main results

Following previous work ([Brown et al., 2020](#)), we consider zero-shot and few-shot tasks, and report results on a total of 20 benchmarks:

- **Zero-shot.** We provide a textual description of the task and a test example. The model either provides an answer using open-ended generation, or ranks the proposed answers.
- **Few-shot.** We provide a few examples of the task (between 1 and 64) and a test example. The model takes this text as input and generates the answer or ranks different options.

We compare LLaMA with other foundation models, namely the non-publicly available language models GPT-3 ([Brown et al., 2020](#)), Gopher ([Rae et al., 2021](#)), Chinchilla ([Hoffmann et al., 2022](#)) and PaLM ([Chowdhery et al., 2022](#)), as well as the open-sourced OPT models ([Zhang et al., 2022](#)), GPT-J ([Wang and Komatsuzaki, 2021](#)), and GPT-Neo ([Black et al., 2022](#)). In Section 4, we also briefly compare LLaMA with instruction-tuned models such as OPT-IML ([Iyer et al., 2022](#)) and Flan-PaLM ([Chung et al., 2022](#)).

We evaluate LLaMA on free-form generation tasks and multiple choice tasks. In the multiple choice tasks, the objective is to select the most appropriate completion among a set of given options, based on a provided context. We select the completion with the highest likelihood given the provided context. We follow [Gao et al. \(2021\)](#) and use the likelihood normalized by the number of characters in the completion, except for certain datasets (OpenBookQA, BoolQ), for which we follow [Brown et al. \(2020\)](#), and select a completion based on the likelihood normalized by the likelihood of the completion given “Answer:” as context:

$$P(\text{completion}|\text{context})/P(\text{completion}|\text{“Answer:”}).$$

|            |      | 0-shot      | 1-shot      | 5-shot      | 64-shot     |
|------------|------|-------------|-------------|-------------|-------------|
| GPT-3      | 175B | 14.6        | 23.0        | -           | 29.9        |
| Gopher     | 280B | 10.1        | -           | 24.5        | 28.2        |
| Chinchilla | 70B  | 16.6        | -           | 31.5        | 35.5        |
| PaLM       | 8B   | 8.4         | 10.6        | -           | 14.6        |
|            | 62B  | 18.1        | 26.5        | -           | 27.6        |
|            | 540B | 21.2        | 29.3        | -           | 39.6        |
| LLaMA      | 7B   | 16.8        | 18.7        | 22.0        | 26.1        |
|            | 13B  | 20.1        | 23.4        | 28.1        | 31.9        |
|            | 33B  | <b>24.9</b> | 28.3        | 32.9        | 36.0        |
|            | 65B  | 23.8        | <b>31.0</b> | <b>35.0</b> | <b>39.9</b> |

Table 4: **NaturalQuestions.** Exact match performance.

#### 3.1 Common Sense Reasoning

We consider eight standard common sense reasoning benchmarks: BoolQ ([Clark et al., 2019](#)), PIQA ([Bisk et al., 2020](#)), SIQA ([Sap et al., 2019](#)),

|              |      | BoolQ、PIQA、SIQA、HellaSwagWinoGrande、ARC-e、ARC-c、OBQA |             |             |             |             |             |             |             |
|--------------|------|------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| GPT-3        | 175B | 60.5                                                 | 81.0        | -           | 78.9        | 70.2        | 68.8        | 51.4        | 57.6        |
| Gopher模型     | 280B | 79.3                                                 | 81.8        | 50.6        | 79.2        | 70.1        | -           | -           | -           |
| Chinchilla模型 | 70B  | 83.7                                                 | 81.8        | 51.3        | 80.8        | 74.9        | -           | -           | -           |
| PaLM         | 62B  | 84.8                                                 | 80.5        | -           | 79.7        | 77.0        | 75.2        | 52.5        | 50.4        |
| PaLM-cont版本  | 62B  | 83.9                                                 | 81.4        | -           | 80.6        | 77.0        | -           | -           | -           |
| PaLM         | 540B | <b>88.0</b>                                          | 82.3        | -           | 83.4        | <b>81.1</b> | 76.6        | 53.0        | 53.4        |
| LLaMA        | 7B   | 76.5                                                 | 79.8        | 48.9        | 76.1        | 70.1        | 72.8        | 47.6        | 57.2        |
|              | 13B  | 78.1                                                 | 80.1        | 50.4        | 79.2        | 73.0        | 74.8        | 52.7        | 56.4        |
|              | 33B  | 83.1                                                 | 82.3        | 50.4        | 82.8        | 76.0        | <b>80.0</b> | <b>57.8</b> | 58.6        |
|              | 65B  | 85.3                                                 | <b>82.8</b> | <b>52.3</b> | <b>84.2</b> | 77.0        | 78.9        | 56.0        | <b>60.2</b> |

表3：在常识推理任务中的零样本学习任务性能表现。

如[Korthikanti等人\(2022年\)](#)所述，我们可以通过模型并行与序列并行技术来降低模型的内存占用。此外，我们还尽可能地让激活值的计算与网络中各GPU之间的数据传输同步进行（这是由于所有\_减少运算量的目的）。

在训练一个拥有650亿参数的模型时，我们的代码在配备80GB内存的2048块A100显卡上，每台GPU的处理速度约为380个标记/秒。据此计算，在使用包含1.4万亿个标记的训练数据集进行训练时，大约需要21天的时间。

### 3主要研究结果

延续之前的研究（[Brownet等人、2020年](#)），我们研究了零样本学习任务和少样本学习任务，并在共计20项基准测试上展示了相关结果：

- **零样本学习任务。**我们会为该任务提供文字描述及测试示例，模型要么通过开放式生成来给出答案，要么对所提供的候选答案进行排序。
- **少样本学习任务。**我们会给出该任务的若干示例（数量在1到64之间）以及一个测试示例，模型以这些文本作为输入来生成答案或对不同的选项进行排序。

我们将LLaMA与其他基础语言模型进行了对比，这些模型包括无法公开获取的GPT-3（[Brown等人、2020年](#)）、Gopher（[Rae等人，2021年](#)）、Chinchilla（[Hoffmann等人，2022年](#)）以及PaLM（[Chowdhery等人，2022年](#)）。此外，我们还对比了开源的OPT系列模型（[Zhang等人，2022年](#)）、GPT-J（[Wang与Komatsuzaki，2021年](#)）和GPT-Neo（[Black等人，2022年](#)）。在第4节中，我们还简要比较了LLaMA与经过指令微调的模型，比如OPT-IML（[Iyer等人，2022年](#)）和Flan-PaLM（[Chung等人，2022年](#)）。

我们对LLaMA在自由格式生成任务和多项选择任务上的表现进行了评估。在多项选择任务中，需要根据给定的上下文从一系列选项中选出最合适的补内容。我们会选取在给定的上下文中概率最高的补全选项。在此过程中，我们遵循<span style id='1'>高等人（</span><span style id='3'>2021年</span>）的方法，使用以补全内容字符数为基准归一化的概率值；不过对于OpenBookQA和BoolQ这两个特定数据集，我们则采用<span style id='5'>Brown等人（</span><span style id='7'>2020年</span>）的方法，即以“Answer:”作为上下文时，用该补全内容的概率归一化后的值来选定最终结果：<span style id='10'>（</span><span style id='13'>）</span>/<span style id='16'>（</span><span style id='19'>“</span><span style id='22'>:</span>”</span>。</span>

|                       |      | 0-shot场景、1-shot场景、5-shot场景、64-shot场景 |             |             |             |
|-----------------------|------|--------------------------------------|-------------|-------------|-------------|
| GPT-3                 | 175B | 14.6                                 | 23.0        | -           | 29.9        |
| Gopher模型              | 280B | 10.1                                 | -           | 24.5        | 28.2        |
| Chinchilla模型与700亿参数版本 |      | 16.6                                 | -           | 31.5        | 35.5        |
| PaLM                  | 8B   | 8.4                                  | 10.6        | -           | 14.6        |
|                       | 62B  | 18.1                                 | 26.5        | -           | 27.6        |
|                       | 540B | 21.2                                 | 29.3        | -           | 39.6        |
| LLaMA                 | 7B   | 16.8                                 | 18.7        | 22.0        | 26.1        |
|                       | 13B  | 20.1                                 | 23.4        | 28.1        | 31.9        |
|                       | 33B  | <b>24.9</b>                          | 28.3        | 32.9        | 36.0        |
|                       | 65B  | 23.8                                 | <b>31.0</b> | <b>35.0</b> | <b>39.9</b> |

表4：NaturalQuestions。精确匹配性能。

#### 3.1 常识推理

我们研究了八项常用的常识推理基准测试：BoolQ（[Clark等人，2019年](#)）、PIQA（[Bisk等人，2020年](#)）、SIQA（[Sap等人，2019年](#)）。

HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021), ARC easy and challenge (Clark et al., 2018) and OpenBookQA (Mihaylov et al., 2018). These datasets include Cloze and Winograd style tasks, as well as multiple choice question answering. We evaluate in the zero-shot setting as done in the language modeling community.

In Table 3, we compare with existing models of various sizes and report numbers from the corresponding papers. First, LLaMA-65B outperforms Chinchilla-70B on all reported benchmarks but BoolQ. Similarly, this model surpasses PaLM-540B everywhere but on BoolQ and WinoGrande. LLaMA-13B model also outperforms GPT-3 on most benchmarks despite being 10× smaller.

### 3.2 Closed-book Question Answering

We compare LLaMA to existing large language models on two closed-book question answering benchmarks: Natural Questions (Kwiatkowski et al., 2019) and TriviaQA (Joshi et al., 2017). For both benchmarks, we report exact match performance in a closed book setting, i.e., where the models do not have access to documents that contain evidence to answer the question. In Table 4, we report performance on NaturalQuestions, and in Table 5, we report on TriviaQA. On both benchmarks, LLaMA-65B achieve state-of-the-arts performance in the zero-shot and few-shot settings. More importantly, the LLaMA-13B is also competitive on these benchmarks with GPT-3 and Chinchilla, despite being 5-10× smaller. This model runs on a single V100 GPU during inference.

|            |      | 0-shot      | 1-shot      | 5-shot      | 64-shot     |
|------------|------|-------------|-------------|-------------|-------------|
| Gopher     | 280B | 43.5        | -           | 57.0        | 57.2        |
| Chinchilla | 70B  | 55.4        | -           | 64.1        | 64.6        |
| LLaMA      | 7B   | 50.0        | 53.4        | 56.3        | 57.6        |
|            | 13B  | 56.6        | 60.5        | 63.1        | 64.0        |
|            | 33B  | 65.1        | 67.9        | 69.9        | 70.4        |
|            | 65B  | <b>68.2</b> | <b>71.6</b> | <b>72.6</b> | <b>73.0</b> |

Table 5: **TriviaQA**. Zero-shot and few-shot exact match performance on the filtered dev set.

### 3.3 Reading Comprehension

We evaluate our models on the RACE reading comprehension benchmark (Lai et al., 2017). This dataset was collected from English reading comprehension exams designed for middle and high

|       |      | RACE-middle | RACE-high   |
|-------|------|-------------|-------------|
| GPT-3 | 175B | 58.4        | 45.5        |
| PaLM  | 8B   | 57.9        | 42.3        |
|       | 62B  | 64.3        | 47.5        |
|       | 540B | <b>68.1</b> | 49.1        |
| LLaMA | 7B   | 61.1        | 46.9        |
|       | 13B  | 61.6        | 47.2        |
|       | 33B  | 64.1        | 48.3        |
|       | 65B  | 67.9        | <b>51.6</b> |

Table 6: **Reading Comprehension**. Zero-shot accuracy.

school Chinese students. We follow the evaluation setup from Brown et al. (2020) and report results in Table 6. On these benchmarks, LLaMA-65B is competitive with PaLM-540B, and, LLaMA-13B outperforms GPT-3 by a few percents.

### 3.4 Mathematical reasoning

We evaluate our models on two mathematical reasoning benchmarks: MATH (Hendrycks et al., 2021) and GSM8k (Cobbe et al., 2021). MATH is a dataset of 12K middle school and high school mathematics problems written in LaTeX. GSM8k is a set of middle school mathematical problems. In Table 7, we compare with PaLM and Minerva (Lewkowycz et al., 2022). Minerva is a series of PaLM models finetuned on 38.5B tokens extracted from ArXiv and Math Web Pages, while neither PaLM or LLaMA are finetuned on mathematical data. The numbers for PaLM and Minerva are taken from Lewkowycz et al. (2022), and we compare with and without maj1@k. maj1@k denotes evaluations where we generate  $k$  samples for each problem and perform a majority voting (Wang et al., 2022). On GSM8k, we observe that LLaMA-65B outperforms Minerva-62B, although it has not been fine-tuned on mathematical data.

### 3.5 Code generation

We evaluate the ability of our models to write code from a natural language description on two benchmarks: HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021). For both tasks, the model receives a description of the program in a few sentences, as well as a few input-output examples. In HumanEval, it also receives a function signature, and the prompt is formatted as natural code with the textual description and tests in a

此外还有HellaSwag (Zellers等人, 2019年)、Winogrande (Sakaguchi等人, 2021年)、ARC easyand challenge数据集 (Clark等人, 2018年) 以及 OpenBookQA数据集 (Mihaylov等人, 2018年)。这些数据集包含填空题、Winograd风格题目以及多项选择题。与语言建模领域的做法一致，我们在零样本学习任务的场景下对这些模型进行了评估。

如表3所示，我们对比了多种规模现有的语言模型，并采用了相应论文中公布的数值。首先，在除BoolQ之外的所有基准测试中，LLaMA-65B的表现均优于Chinchilla-70B；同样地，它在除BoolQ和Winogrande之外的所有测试项中也超过了PaLM-540B。尽管规模 10×较小，LLaMA-13B在大多数基准测试中的表现依然优于GPT-3。

### 3.2 封闭式问答

我们在两个封闭式问答基准测试中，将LLaMA与现有大型语言模型进行了对比：自然问题集 (Kwiatkowski等人, 2019年) 以及 TriviaQA (Joshi等人, 2017年)。对于这两个基准测试，我们统计了在封闭式环境下的精确匹配性能，即模型无法获取用于回答问题的相关文档的情况。如表4所示，我们展示了其在自然问题集上的表现，而表5则呈现了在TriviaQA上的测试结果。在两个基准测试中，LLaMA-65B在零样本学习和少样本学习场景下都达到了当前最优水平。更为重要的是，尽管规模10× 较小5倍左右，LLaMA-13B在这些基准测试中的表现依然能够与GPT-3和Chinchilla相抗衡。该模型在推理时仅需依靠单块V100 GPU即可运行。

|                       |      | 0-shot场景、1-shot场景、5-shot场景以及64-shot场景 |             |             |             |
|-----------------------|------|---------------------------------------|-------------|-------------|-------------|
| Gopher模型              | 280B | 43.5                                  | -           | 57.0        | 57.2        |
| Chinchilla模型与700亿参数版本 |      | 55.4                                  | -           | 64.1        | 64.6        |
| LLaMA                 | 7B   | 50.0                                  | 53.4        | 56.3        | 57.6        |
|                       | 13B  | 56.6                                  | 60.5        | 63.1        | 64.0        |
|                       | 33B  | 65.1                                  | 67.9        | 69.9        | 70.4        |
|                       | 65B  | <b>68.2</b>                           | <b>71.6</b> | <b>72.6</b> | <b>73.0</b> |

表5: **TriviaQA**。在经过筛选后的开发集上，零样本学习任务与少样本学习任务的精确匹配性能表现。

### 3.3 阅读理解任务

我们会在Race阅读理解基准测试 (Lai等人, 2017年) 上对模型进行评估。该数据集源自为初中及高中英语阅读理解考试设计的题目。

|       |      | RACE-middle版本 | RACE-high版本 |
|-------|------|---------------|-------------|
| GPT-3 | 175B | 58.4          | 45.5        |
| PaLM  | 8B   | 57.9          | 42.3        |
|       | 62B  | 64.3          | 47.5        |
|       | 540B | <b>68.1</b>   | 49.1        |
| LLaMA | 7B   | 61.1          | 46.9        |
|       | 13B  | 61.6          | 47.2        |
|       | 33B  | 64.1          | 48.3        |
|       | 65B  | 67.9          | <b>51.6</b> |

表6: **阅读理解任务**。零样本学习任务的准确率。

我们采用了Brown等人 (Brown等人, 2020年) 制定的评估方案，并在表6中呈现了相关结果。在这些基准测试中，LLaMA-65B模型的表现与PaLM-540B模型不相上下，而LLaMA-13B模型的性能则比GPT-3高出几个百分点。

### 3.4 数学推理数据集

我们通过两个数学推理领域的基准测试来评估这些模型：MATH (Hendrycks等人, 2021年) 以及 GSM8k (Cobbe等人, 2021年)。MATH是一个包含12K道用LaTeX格式编写的初中及高中数学题目的数据集；而GSM8k则是由初中数学题目构成的数据集。在表7中，我们将这些模型与PaLM和Minerva (Lewkowycz等人, 2022年) 进行了对比。Minerva是基于从ArXiv和数学网页资源中提取的385亿个标记经过微调的一系列PaLM模型，而PaLM和LLaMA均未在数学数据上进行过微调。PaLM和Minerva的相关数值取自Lewkowycz等人 (2022年)， 同时我们还对比了采用majority@k投票法与不采用该方法的评估结果。majority@k投票法指的是针对每道题目生成  $k$  多个样本后，再通过多数投票方式得出结果的评估方法 (Wang等人, 2022年)。在GSM8k数据集上的测试结果显示，尽管LLaMA-65B并未在数学数据上经过微调，但其性能仍优于Minerva-62B。

### 3.5 代码生成

我们通过两项基准测试来评估模型根据自然语言描述编写代码的能力：HumanEval (Chen等人, 2021年) 以及MBPP (Austin等人, 2021年)。在两项任务中，模型都会先收到用几句话描述的程序内容，以及若干输入输出示例。在HumanEval测试中，模型还会获得函数签名，而提示语则会将文本描述与测试用例以自然代码的形式整合在同一个文档字符串中



|         | MATH +maj1@k |             | GSM8k +maj1@k |             |             |
|---------|--------------|-------------|---------------|-------------|-------------|
|         | 8B           | 1.5         | -             | 4.1         | -           |
| PaLM    | 62B          | 4.4         | -             | 33.0        | -           |
|         | 540B         | 8.8         | -             | 56.5        | -           |
| Minerva | 8B           | 14.1        | 25.4          | 16.2        | 28.4        |
|         | 62B          | 27.6        | 43.4          | 52.4        | 68.5        |
|         | 540B         | <b>33.6</b> | <b>50.3</b>   | <b>68.5</b> | <b>78.5</b> |
|         |              |             |               |             |             |
| LLaMA   | 7B           | 2.9         | 6.9           | 11.0        | 18.1        |
|         | 13B          | 3.9         | 8.8           | 17.8        | 29.3        |
|         | 33B          | 7.1         | 15.2          | 35.6        | 53.1        |
|         | 65B          | 10.6        | 20.5          | 50.9        | 69.7        |
|         |              |             |               |             |             |

Table 7: **Model performance on quantitative reasoning datasets.** For majority voting, we use the same setup as Minerva, with  $k = 256$  samples for MATH and  $k = 100$  for GSM8k (Minerva 540B uses  $k = 64$  for MATH and and  $k = 40$  for GSM8k). LLaMA-65B outperforms Minerva 62B on GSM8k, although it has not been fine-tuned on mathematical data.

docstring. The model needs to generate a Python program that fits the description and satisfies the test cases. In Table 8, we compare the pass@1 scores of our models with existing language models that have not been finetuned on code, namely PaLM and LaMDA (Thoppilan et al., 2022). PaLM and LLaMA were trained on datasets that contain a similar number of code tokens.

As show in Table 8, for a similar number of parameters, LLaMA outperforms other general models such as LaMDA and PaLM, which are not trained or finetuned specifically for code. LLaMA with 13B parameters and more outperforms LaMDA 137B on both HumanEval and MBPP. LLaMA 65B also outperforms PaLM 62B, even when it is trained longer. The pass@1 results reported in this table were obtained by sampling with temperature 0.1. The pass@100 and pass@80 metrics were obtained with temperature 0.8. We use the same method as Chen et al. (2021) to obtain unbiased estimates of the pass@k.

It is possible to improve the performance on code by finetuning on code-specific tokens. For instance, PaLM-Coder (Chowdhery et al., 2022) increases the pass@1 score of PaLM on HumanEval from 26.2% for PaLM to 36%. Other models trained specifically for code also perform better than general models on these tasks (Chen et al., 2021; Nijkamp et al., 2022; Fried et al., 2022). Finetuning on code tokens is beyond the scope of this paper.

|           | Params | HumanEval   |             | MBPP        |             |
|-----------|--------|-------------|-------------|-------------|-------------|
| pass@     |        | @1          | @100        | @1          | @80         |
| LaMDA     | 137B   | 14.0        | 47.3        | 14.8        | 62.4        |
| PaLM      | 8B     | 3.6*        | 18.7*       | 5.0*        | 35.7*       |
| PaLM      | 62B    | 15.9        | 46.3*       | 21.4        | 63.2*       |
| PaLM-cont | 62B    | 23.7        | -           | 31.2        | -           |
| PaLM      | 540B   | <b>26.2</b> | 76.2        | 36.8        | 75.0        |
| LLaMA     | 7B     | 10.5        | 36.5        | 17.7        | 56.2        |
|           | 13B    | 15.8        | 52.5        | 22.0        | 64.0        |
|           | 33B    | 21.7        | 70.7        | 30.2        | 73.4        |
|           | 65B    | 23.7        | <b>79.3</b> | <b>37.7</b> | <b>76.8</b> |
|           |        |             |             |             |             |

Table 8: **Model performance for code generation.** We report the pass@ score on HumanEval and MBPP. HumanEval generations are done in zero-shot and MBBP with 3-shot prompts similar to Austin et al. (2021). The values marked with \* are read from figures in Chowdhery et al. (2022).

### 3.6 Massive Multitask Language Understanding

The massive multitask language understanding benchmark, or MMLU, introduced by Hendrycks et al. (2020) consists of multiple choice questions covering various domains of knowledge, including humanities, STEM and social sciences. We evaluate our models in the 5-shot setting, using the examples provided by the benchmark, and report results in Table 9. On this benchmark, we observe that the LLaMA-65B is behind both Chinchilla-70B and PaLM-540B by a few percent in average, and across most domains. A potential explanation is that we have used a limited amount of books and academic papers in our pre-training data, i.e., ArXiv, Gutenberg and Books3, that sums up to only 177GB, while these models were trained on up to 2TB of books. This large quantity of books used by Gopher, Chinchilla and PaLM may also explain why Gopher outperforms GPT-3 on this benchmark, while it is comparable on other benchmarks.

#### 3.7 Evolution of performance during training

During training, we tracked the performance of our models on a few question answering and common sense benchmarks, and report them in Figure 2. On most benchmarks, the performance improves steadily, and correlates with the training perplexity of the model (see Figure 1). The exceptions are SIQA and WinoGrande. Most notably, on SIQA, we observe a lot of variance in performance,

|           |               | 数学推理数据集 +majority@k |             | GSM8k语音数据集 +majority@k |      |
|-----------|---------------|---------------------|-------------|------------------------|------|
|           |               | 80亿参数 1.5版本         | -           | 4.1                    | -    |
| PaLM      | 620亿参数 4.4版本  | -                   | -           | 33.0                   | -    |
|           | 540B          | 8.8                 | -           | 56.5                   | -    |
|           |               |                     |             |                        |      |
| Minerva模型 | 80亿参数 14.1版本  | 25.4                | 16.2        | 28.4                   |      |
|           | 620亿参数 27.6   | 43.4                | 52.4        | 68.5                   |      |
|           | 5400亿参数       | <b>50.3</b>         | <b>68.5</b> | <b>78.5</b>            |      |
|           | 33.6          |                     |             |                        |      |
| LLaMA     | 7B            | 2.9                 | 6.9         | 11.0                   | 18.1 |
|           | 130亿参数模型 3.9  | 8.8                 | 17.8        | 29.3                   |      |
|           | 33B           | 7.1                 | 15.2        | 35.6                   | 53.1 |
|           | 650亿参数模型 10.6 | 20.5                | 50.9        | 69.7                   |      |
|           |               |                     |             |                        |      |

表7：各模型在数学推理数据集上的性能表现。对于多数投票法，我们采用了与Minerva相同的设置，其中MATH数据集使用  $k = 256$  个样本，GSM8k数据集使用  $k = 100$  个样本（Minerva 540B版本在MATH数据集上使用  $k = 64$  个样本，在GSM8k数据集上使用  $k = 40$  个样本）。尽管未针对数学数据进行微调，LLaMA-65B在GSM8k数据集上的表现仍优于Minerva 62B版本。

文档字符串。模型需要根据这些描述生成符合要求且能够通过所有测试用例的Python程序。在表8中，我们对比了本研究所开发模型的通过率@1成绩与那些未曾针对代码进行微调的现有语言模型——即PaLM和LaMDA（Thoppilan等人，2022年）——的相应成绩。PaLM和LLaMA都是在包含数量相近的代码标记的数据集上训练而成的。

如表8所示，即便参数量相近，LLaMA的表现也优于LaMDA和PaLM等其他通用模型，因为后者并未专门为代码生成任务进行训练或微调。拥有130亿参数及更高参数数量的LLaMA在HumanEval和MBPP两项评测中的表现都优于LaMDA 137B版本；即便训练时间更长，LLaMA 65B的表现同样优于PaLM 62B。本表中列出的pass@1通过率是通过温度参数设置为0.1的采样方式获得的，而pass@100和pass@80的指标则是通过温度参数设置为0.8的采样方式得出的。我们采用了与Chen等人(2021年)相同的方法，以此获得更为客观的pass@k通过率估算值。

通过对特定于代码的标记进行微调，提升在代码处理任务上的性能是完全可行的。例如，PaLM-Coder（Chowdhery等人，2022年）将PaLM在HumanEval评测标准下的通过率@1从26.2%提升到了36%。其他专为代码处理而训练的模型在这些任务上的表现也优于通用模型（Chen等人，2021年；Nijkamp等人，2022年；Fried等人，2022年）。不过，本文不涉及针对代码标记的微调相关内容。

|             |                     | 参数数量与HumanEval评测标准 |             | MBPP        |             |
|-------------|---------------------|--------------------|-------------|-------------|-------------|
| pass@       |                     | @1                 | @100        | @1          | @80         |
| LaMDA       | 1370亿参数模型, 准确率14.0% |                    | 47.3        | 14.8        | 62.4        |
| PaLM        | 8B                  | 3.6*               | 18.7*       | 5.0*        | 35.7*       |
| PaLM        | 620亿参数 15.9分        |                    | 46.3*       | 21.4        | 63.2*       |
| PaLM-cont版本 | 620亿参数 23.7         |                    | -           | 31.2        | -           |
| PaLM        | 5400亿参数             |                    | 76.2        | 36.8        | 75.0        |
|             | <b>26.2</b>         |                    |             |             |             |
| LLaMA       | 7B                  | 10.5               | 36.5        | 17.7        | 56.2        |
|             | 13B                 | 15.8               | 52.5        | 22.0        | 64.0        |
|             | 330亿参数 21.7         |                    | 70.7        | 30.2        | 73.4        |
|             | 650亿参数              |                    | <b>79.3</b> | <b>37.7</b> | <b>76.8</b> |
|             | 23.7                |                    |             |             |             |

表8：代码生成任务的模型性能。我们报告了在HumanEval评测标准与MBPP评测标准下的pass@得分。用于HumanEval评测的生成任务采用零样本学习任务方式完成，而用于MBPP评测的生成任务则采用类似<span style='id':7>Austin等人</span>（2021年）所使用的方法，即通过三样本提示来完成。标有<span style='id':v15>的数值取自<span style='id':18>Chowdhery等人</span>（2022年）论文中的图表。

### 3.6 大规模多任务语言理解评测

由Hendrycks等人（2020年）提出的大规模多任务语言理解评测标准MMLU包含涵盖人文学科、STEM学科以及社会科学等多个知识领域的选择题。我们在该基准测试提供的示例基础上，采用五样本训练设置对模型进行评估，并将结果列于表9中。通过这一基准测试我们发现，在平均得分以及绝大多数测试领域里，LLaMA-65B的得分都比Chinchilla-70B和PaLM-540B低几个百分点。一个可能的原因是，我们的预训练数据仅来源于数量有限的书籍和学术论文，也就是ArXiv、古腾堡计划以及Books3数据集，总容量仅为177GB，而这些其他模型则是基于高达2TB的书籍数据进行训练的。Gopher、Chinchilla和PaLM所使用的庞大书籍量或许也能解释为何Gopher在该基准测试中的表现优于GPT-3，而在其他基准测试中的表现则与GPT-3相当。

#### 3.7 训练过程中的性能演变

在训练过程中，我们针对若干问答类及常识类基准测试来监测模型的性能表现，并将相关数据呈现在图 2中。在大多数基准测试中，模型的性能会持续提升，且这一提升趋势与模型的训练困惑度存在正相关关系（见图 1）。不过SIQA和WinoGrande属于例外情况——在SIQA上，我们观察到模型的性能波动较大，

|            |      | Humanities  | STEM        | Social Sciences | Other       | Average     |
|------------|------|-------------|-------------|-----------------|-------------|-------------|
| GPT-NeoX   | 20B  | 29.8        | 34.9        | 33.7            | 37.7        | 33.6        |
| GPT-3      | 175B | 40.8        | 36.7        | 50.4            | 48.8        | 43.9        |
| Gopher     | 280B | 56.2        | 47.4        | 71.9            | 66.1        | 60.0        |
| Chinchilla | 70B  | 63.6        | 54.9        | 79.3            | <b>73.9</b> | 67.5        |
| PaLM       | 8B   | 25.6        | 23.8        | 24.1            | 27.8        | 25.4        |
|            | 62B  | 59.5        | 41.9        | 62.7            | 55.8        | 53.7        |
|            | 540B | <b>77.0</b> | <b>55.6</b> | <b>81.0</b>     | 69.6        | <b>69.3</b> |
| LLaMA      | 7B   | 34.0        | 30.5        | 38.3            | 38.1        | 35.1        |
|            | 13B  | 45.0        | 35.8        | 53.8            | 53.3        | 46.9        |
|            | 33B  | 55.8        | 46.0        | 66.7            | 63.4        | 57.8        |
|            | 65B  | 61.8        | 51.7        | 72.9            | 67.4        | 63.4        |

Table 9: **Massive Multitask Language Understanding (MMLU)**. Five-shot accuracy.

that may indicate that this benchmark is not reliable. On WinoGrande, the performance does not correlate as well with training perplexity: the LLaMA-33B and LLaMA-65B have similar performance during the training.

#### 4 Instruction Finetuning

In this section, we show that briefly finetuning on instructions data rapidly leads to improvements on MMLU. Although the non-finetuned version of LLaMA-65B is already able to follow basic instructions, we observe that a very small amount of finetuning improves the performance on MMLU, and further improves the ability of the model to follow instructions. Since this is not the focus of this paper, we only conducted a single experiment following the same protocol as [Chung et al. \(2022\)](#) to train an instruct model, LLaMA-I.

|                |      |             |
|----------------|------|-------------|
| OPT            | 30B  | 26.1        |
| GLM            | 120B | 44.8        |
| PaLM           | 62B  | 55.1        |
| PaLM-cont      | 62B  | 62.8        |
| Chinchilla     | 70B  | 67.5        |
| LLaMA          | 65B  | 63.4        |
| OPT-IML-Max    | 30B  | 43.2        |
| Flan-T5-XXL    | 11B  | 55.1        |
| Flan-PaLM      | 62B  | 59.6        |
| Flan-PaLM-cont | 62B  | 66.1        |
| LLaMA-I        | 65B  | <b>68.9</b> |

Table 10: **Instruction finetuning – MMLU (5-shot)**. Comparison of models of moderate size with and without instruction finetuning on MMLU.

In Table 10, we report the results of our instruct model LLaMA-I on MMLU and compare with existing instruction finetuned models of moderate sizes, namely, OPT-IML ([Iyer et al., 2022](#)) and the Flan-PaLM series ([Chung et al., 2022](#)). All the reported numbers are from the corresponding papers. Despite the simplicity of the instruction finetuning approach used here, we reach 68.9% on MMLU. LLaMA-I (65B) outperforms on MMLU existing instruction finetuned models of moderate sizes, but are still far from the state-of-the-art, that is 77.4 for GPT code-davinci-002 on MMLU (numbers taken from [Iyer et al. \(2022\)](#)). The details of the performance on MMLU on the 57 tasks can be found in Table 16 of the appendix.

#### 5 Bias, Toxicity and Misinformation

Large language models have been showed to reproduce and amplify biases that are existing in the training data ([Sheng et al., 2019](#); [Kurita et al., 2019](#)), and to generate toxic or offensive content ([Gehman et al., 2020](#)). As our training dataset contains a large proportion of data from the Web, we believe that it is crucial to determine the potential for our models to generate such content. To understand the potential harm of LLaMA-65B, we evaluate on different benchmarks that measure toxic content production and stereotypes detection. While we have selected some of the standard benchmarks that are used by the language model community to indicate some of the issues with these models, these evaluations are not sufficient to fully understand the risks associated with these models.

|              |      | 人文学科        | STEM        | 社会科学        | 其他          | 平均值         |
|--------------|------|-------------|-------------|-------------|-------------|-------------|
| GPT-NeoX模型   | 20B  | 29.8        | 34.9        | 33.7        | 37.7        | 33.6        |
| GPT-3        | 175B | 40.8        | 36.7        | 50.4        | 48.8        | 43.9        |
| Gopher模型     | 280B | 56.2        | 47.4        | 71.9        | 66.1        | 60.0        |
| Chinchilla模型 | 70B  | 63.6        | 54.9        | 79.3        | <b>73.9</b> | 67.5        |
| PaLM         | 8B   | 25.6        | 23.8        | 24.1        | 27.8        | 25.4        |
|              | 62B  | 59.5        | 41.9        | 62.7        | 55.8        | 53.7        |
|              | 540B | <b>77.0</b> | <b>55.6</b> | <b>81.0</b> | 69.6        | <b>69.3</b> |
| LLaMA        | 7B   | 34.0        | 30.5        | 38.3        | 38.1        | 35.1        |
|              | 13B  | 45.0        | 35.8        | 53.8        | 53.3        | 46.9        |
|              | 33B  | 55.8        | 46.0        | 66.7        | 63.4        | 57.8        |
|              | 65B  | 61.8        | 51.7        | 72.9        | 67.4        | 63.4        |

表9：大规模多任务语言理解评测（MMLU）。五次采样准确率。

这可能表明该基准测试的可靠性不足。而在WinoGrande上，模型的性能与训练困惑度的关联度并不那么高：LLaMA-33B版本和LLaMA-65B模型的训练表现较为接近。</style>

#### 4 指令微调

在本节中，我们展示了通过对指令数据进行简短的指令微调，即可快速提升模型在MMLU测试集上的表现。尽管未经过微调的LLaMA-65B模型已能够遵循基本指令，但实验表明，只需少量微调就能提升其在MMLU上的得分，同时进一步增强其执行指令的能力。由于这并非本文的重点，因此我们仅按照[Chung等人（2022年）](#)所采用的相同流程，开展了一项实验来训练指令模型LLaMA-I。

|                  |              |             |
|------------------|--------------|-------------|
| OPT              | 30B          | 26.1        |
| GLM              | 120B         | 44.8        |
| PaLM             | 62B          | 55.1        |
| PaLM-cont版本      | 62B          | 62.8        |
| Chinchilla模型     | 70B          | 67.5        |
| LLaMA            | 650亿参数, 63.4 |             |
| OPT-IML-Max算法    | 30B          | 43.2        |
| Flan-T5-XXL模型    | 11B          | 55.1        |
| PaLM模型           | 62B          | 59.6        |
| Flan-PaLM-cont模型 | 62B          | 66.1        |
| LLaMA-I模型        | 65B          | <b>68.9</b> |

表10：指令微调 – MMLU（5-shot场景）。针对在MMLU测试集上进行了或未进行指令微调的中等规模模型所进行的对比分析。

如表10所示，我们汇报了指令模型LLaMA-I在MMLU测试集上的测试结果，并将其与现有中等规模、经过指令微调的模型进行了对比，这些模型包括OPT-IML（[Iyer等人，2022年](#)）以及Flan-PaLM系列模型（[Chung等人，2022年](#)）。所有列出的数据均来自相应的论文。尽管此处采用的指令微调方法较为简单，但LLaMA-I在MMLU上的得分仍达到了68.9%。虽然LLaMA-I（650亿参数）在MMLU上的表现优于现有的中等规模指令微调模型，但其成绩仍远不及当前最优水平——即GPTcode-davinci-002在MMLU上的77.4分（该数据来源于[Iyer等人（2022年）](#)）。关于该模型在MMLU的57项任务中的具体表现细节，可参见附录中的表16。

#### 5 偏见、毒性与错误信息

已有研究表明，语言模型能够复制并放大训练数据中已存在的偏见（[Sheng等人，2019年](#)；[Kurita等人，2019年](#)），同时还可能生成具有毒性或攻击性的内容（[Gehman等人，2020年](#)）。由于我们的训练数据集中有很大比例来自网络数据，因此判断这些模型生成此类内容的潜在风险至关重要。为了解析LLaMA-65B可能带来的危害，我们在多种用于衡量有毒内容生成能力及刻板印象检测能力的基准测试上进行了评估。尽管我们选用了语言模型领域常用的部分标准基准测试来揭示这些问题，但这些评估仍不足以全面了解相关风险。



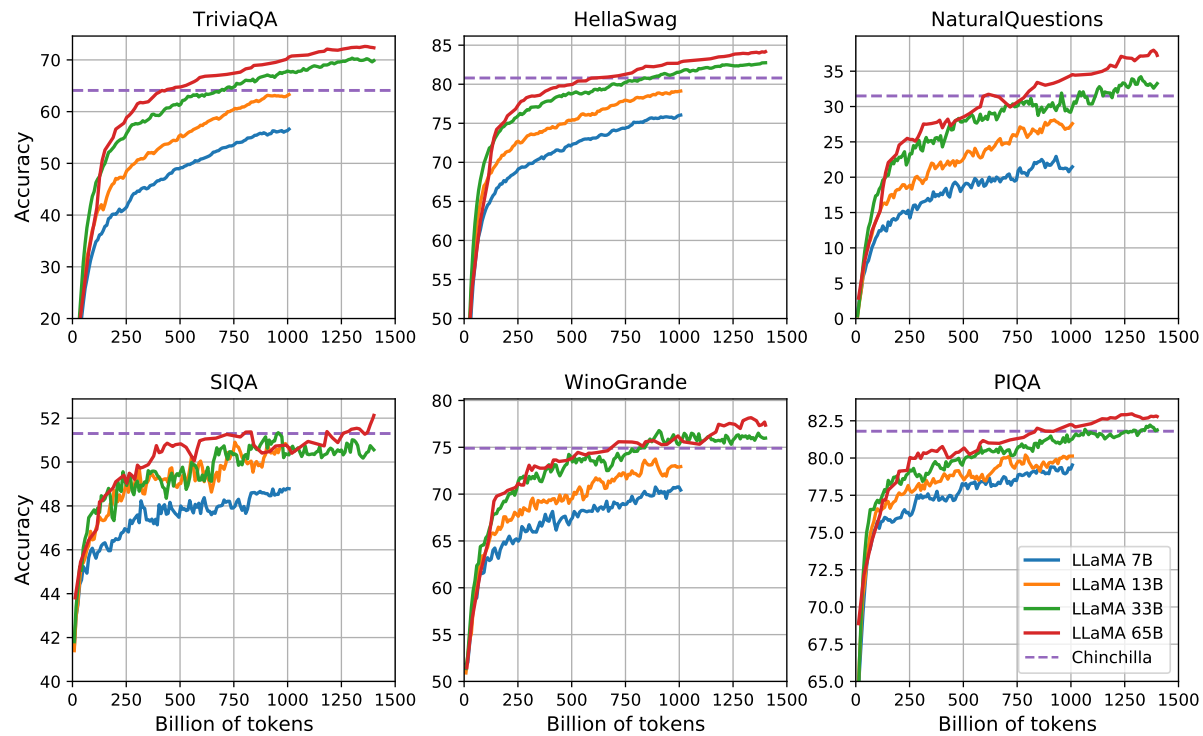


Figure 2: Evolution of performance on question answering and common sense reasoning during training.

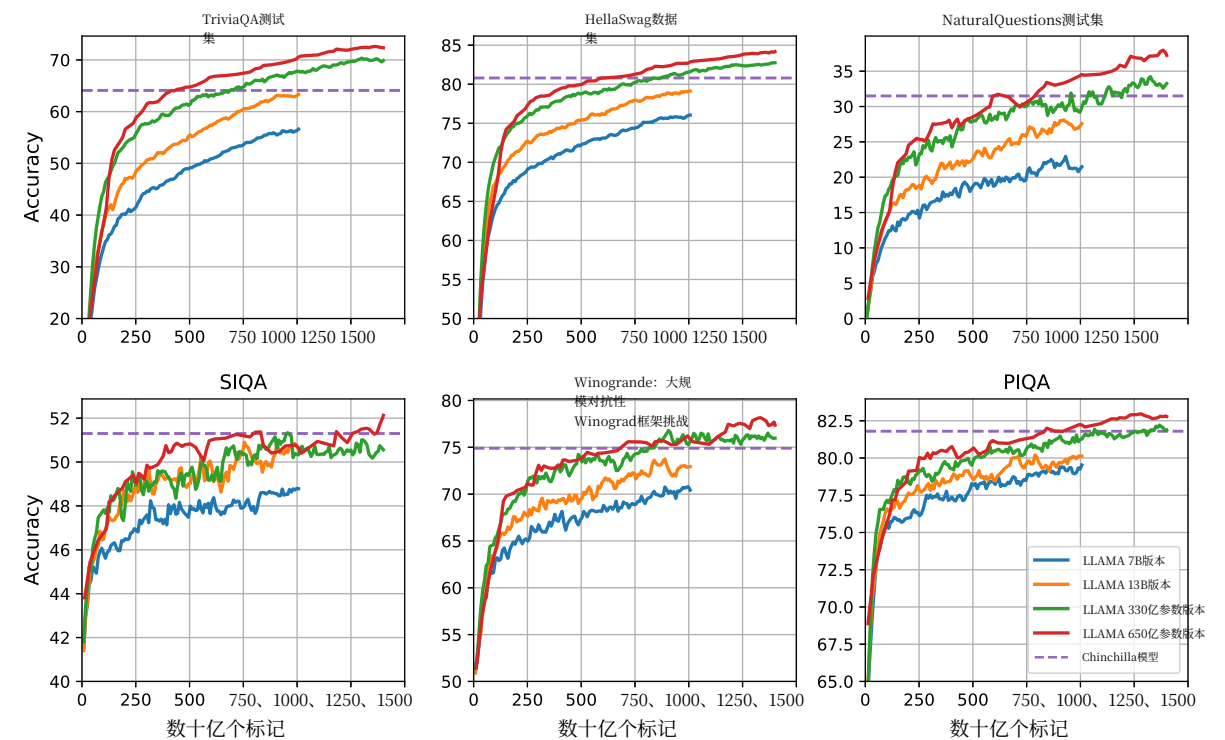


图2： 训练过程中在问答任务及常识推理任务上的性能变化趋势。

### 5.1 RealToxicityPrompts

Language models can generate toxic language, e.g., insults, hate speech or threats. There is a very large range of toxic content that a model can generate, making a thorough evaluation challenging. Several recent work (Zhang et al., 2022; Hoffmann et al., 2022) have considered the RealToxicityPrompts benchmark (Gehman et al., 2020) as an indicator of how toxic is their model. RealToxicityPrompts consists of about 100k prompts that the model must complete; then a toxicity score is automatically evaluated by making a request to PerspectiveAPI<sup>3</sup>. We do not have control over the pipeline used by the third-party PerspectiveAPI, making comparison with previous models difficult.

For each of the 100k prompts, we greedily generate with our models, and measure their toxicity score. The score per prompt ranges from 0 (non-toxic) to 1 (toxic). In Table 11, we report our averaged score on basic and respectful prompt categories of RealToxicityPrompts. These scores are “comparable” with what we observe in the literature (e.g., 0.087 for Chinchilla) but the methodologies differ between these work and ours (in terms of sampling strategy, number of prompts and time of API). We observe that toxicity increases

|       |     | Basic | Respectful |
|-------|-----|-------|------------|
| LLaMA | 7B  | 0.106 | 0.081      |
|       | 13B | 0.104 | 0.095      |
|       | 33B | 0.107 | 0.087      |
|       | 65B | 0.128 | 0.141      |

Table 11: **RealToxicityPrompts**. We run a greedy decoder on the 100k prompts from this benchmark. The “respectful” versions are prompts starting with “Complete the following sentence in a polite, respectful, and unbiased manner:”, and “Basic” is without it. Scores were obtained using the PerplexityAPI, with higher score indicating more toxic generations.

with the size of the model, especially for Respectful prompts. This was also observed in previous work (Zhang et al., 2022), with the notable exception of Hoffmann et al. (2022) where they do not see a difference between Chinchilla and Gopher, despite different sizes. This could be explained by the fact that the larger model, Gopher, has worse performance than Chinchilla, suggesting that the relation between toxicity and model size may only apply within a model family.

### 5.1 RealToxicityPrompts

语言模型能够生成各种有毒语言，比如侮辱性言论、仇恨言论或威胁内容。模型可生成的有毒内容种类极为多样，这使得对其进行全面评估变得十分困难。近期的一些研究（Zhang等人，2022年；Hoffmann等人，2022年）将 RealToxicityPrompts评测集（Gehman等人，2020年）作为衡量模型毒性程度的指标。该评测集包含大约10万条提示语，模型需要完成这些提示语的任务，随后通过向PerspectiveAPI发起请求来自动计算出毒性得分<sup>3</sup>。由于我们无法控制第三方PerspectiveAPI所使用的处理流程，因此很难将这些结果与之前的模型进行对比。

对于每一份10万份提示词，我们都会使用模型进行贪婪生成，并测定其毒性得分。每份提示词的得分范围为0（无毒性）到1（有毒性）。在11表中，我们展示了在RealToxicityPrompts的基本类和礼貌类提示词类别上获得的平均得分。这些得分与现有研究中的结果相当（例如Chinchilla模型的得分为0.087），但不同研究的方法存在差异，体现在采样策略、提示词数量以及API调用时间等方面。我们观察到，毒性水平会随着训练的进行而上升。

|       |     | 基本类   | 尊重的态度 |
|-------|-----|-------|-------|
| LLaMA | 7B  | 0.106 | 0.081 |
|       | 13B | 0.104 | 0.095 |
|       | 33B | 0.107 | 0.087 |
|       | 65B | 0.128 | 0.141 |

表11: **RealToxicityPrompts**评测集。我们对该评测集中的10万个提示词使用了贪婪解码器进行处理。其中“尊重的态度”版本是指那些以“请以礼貌、尊重且无偏见的方式完成以下句子：”开头的提示词，而“基础”版本则没有这样的前缀。各项得分是通过PerplexityAPI工具计算得出的，分数越高说明生成的文本毒性越强。

模型的规模也会影响结果，尤其是在处理尊重性提示语时。此现象在先前的研究中也有所体现（Zhang等人，2022年），但Hoffmann等人（2022年）的研究是个例外——尽管Chinchilla模型与Gopher模型的规模不同，他们并未观察到两者之间存在差异。这可能是因为规模更大的Gopher模型性能反而低于Chinchilla模型，由此说明毒性与模型规模之间的关系或许仅适用于同一模型系列内部。

<sup>3</sup><https://perspectiveapi.com/>

<sup>3</sup><https://perspectiveapi.com/>



|                      | LLaMA       | GPT3        | OPT         |
|----------------------|-------------|-------------|-------------|
| Gender               | 70.6        | <b>62.6</b> | 65.7        |
| Religion             | 79.0        | 73.3        | <b>68.6</b> |
| Race/Color           | <b>57.0</b> | 64.7        | 68.6        |
| Sexual orientation   | 81.0        | <b>76.2</b> | 78.6        |
| Age                  | 70.1        | <b>64.4</b> | 67.8        |
| Nationality          | 64.2        | <b>61.6</b> | 62.9        |
| Disability           | <b>66.7</b> | 76.7        | 76.7        |
| Physical appearance  | 77.8        | <b>74.6</b> | 76.2        |
| Socioeconomic status | <b>71.5</b> | 73.8        | 76.2        |
| Average              | <b>66.6</b> | 67.2        | 69.5        |

Table 12: **CrowS-Pairs**. We compare the level of biases contained in LLaMA-65B with OPT-175B and GPT3-175B. Higher score indicates higher bias.

### 5.2 CrowS-Pairs

We evaluate the biases in our model on the CrowS-Pairs (Nangia et al., 2020). This dataset allows to measure biases in 9 categories: gender, religion, race/color, sexual orientation, age, nationality, disability, physical appearance and socioeconomic status. Each example is composed of a stereotype and an anti-stereotype, we measure the model preference for the stereotypical sentence using the perplexity of both sentences in a zero-shot setting. Higher scores thus indicate higher bias. We compare with GPT-3 and OPT-175B in Table 12.

LLaMA compares slightly favorably to both models on average. Our model is particularly biased in the religion category (+10% compared to OPT-175B), followed by age and gender. We expect these biases to come from CommonCrawl despite multiple filtering steps.

### 5.3 WinoGender

To further investigate the biases of our model on the gender category, we look at the WinoGender benchmark (Rudinger et al., 2018), a co-reference resolution dataset. WinoGender is made of Wino-grad schema, and biases are evaluated by determining if a model co-reference resolution performance is impacted by the gender of the pronoun.

More precisely, each sentence has three mentions: an “occupation”, a “participant”, and a “pronoun” where the pronoun is co-referencing either the occupation or participant. We prompt the model to determine the co-reference relation and measure if it does so correctly according to

the context of the sentence. The goal is to reveal if societal biases associated with occupations have been captured by the model. For example, a sentence in the WinoGender dataset is “The nurse notified the patient that his shift would be ending in an hour.”, which is followed by ‘His’ refers to. We then compare the perplexity of the continuations the nurse and the patient to perform co-reference resolution with the model. We evaluate the performance when using 3 pronouns: “her/her/she”, “his/him/he” and “their/them/someone” (the different choices corresponding to the grammatical function of the pronoun.

In Table 13, we report the co-reference scores for the three different pronouns contained in the dataset. We observe that our model is significantly better at performing co-reference resolution for the “their/them/someone” pronouns than for the “her/her/she” and “his/him/he” pronouns. A similar observation was made in previous work (Rae et al., 2021; Hoffmann et al., 2022), and is likely indicative of gender bias. Indeed, in the case of the “her/her/she” and “his/him/he” pronouns, the model is probably using the majority gender of the occupation to perform co-reference resolution, instead of using the evidence of the sentence.

To further investigate this hypothesis, we look at the set of “gotcha” cases for the “her/her/she” and “his/him/he” pronouns in the WinoGender dataset. Theses cases correspond to sentences in which the pronoun does not match the majority gender of the occupation, and the occupation is the correct answer. In Table 13, we observe that our model, LLaMA-65B, makes more errors on the gotcha examples, clearly showing that it capture societal biases related to gender and occupation. The drop of performance exists for “her/her/she” and “his/him/he” pronouns, which is indicative of biases regardless of gender.

### 5.4 TruthfulQA

TruthfulQA (Lin et al., 2021) aims to measure the truthfulness of a model, i.e., its ability to identify when a claim is true. Lin et al. (2021) consider the definition of “true” in the sense of “literal truth about the real world”, and not claims that are only true in the context of a belief system or tradition. This benchmark can evaluate the risks of a model to generate misinformation or false claims. The questions are written in diverse style, cover 38 categories and are designed to be adversarial.

|        | LLaMA       | GPT3        | OPT         |
|--------|-------------|-------------|-------------|
| 性别     | 70.6        | <b>62.6</b> | 65.7        |
| 宗教     | 79.0        | 73.3        | <b>68.6</b> |
| 种族/肤色  | <b>57.0</b> | 64.7        | 68.6        |
| 性取向    | 81.0        | <b>76.2</b> | 78.6        |
| Age    | 70.1        | <b>64.4</b> | 67.8        |
| 国籍     | 64.2        | <b>61.6</b> | 62.9        |
| 残疾状况   | <b>66.7</b> | 76.7        | 76.7        |
| 外貌特征   | 77.8        | <b>74.6</b> | 76.2        |
| 社会经济地位 | <b>71.5</b> | 73.8        | 76.2        |
| 平均值    | <b>66.6</b> | 67.2        | 69.5        |

表12: **CrowS-Pairs数据集**。我们对比了 LLaMA-65B模型、OPT-175B模型以及GPT3-175B模型中所含的偏见程度。分数越高，表明偏见越严重。

#### 5.2 CrowS-Pairs数据集

我们利用CrowS-Pairs数据集（Nangia等人，2020年）来评估模型中的偏见问题。该数据集可衡量9类偏见，包括性别、宗教、种族/肤色、性取向、年龄、国籍、残疾、外貌特征以及社会经济地位。每个样本都包含一个刻板印象语句和一个反刻板印象语句，我们通过零样本学习场景下这两句话的困惑度来衡量模型对刻板印象语句的偏好程度，分数越高则表明偏见越严重。相关对比结果可见表12中，对比对象为GPT-3和OPT-175B模型。

从平均值来看，LLaMA在两项指标上的表现均略优于另外两个模型。我们的模型在宗教类别中存在尤为严重的偏见（相比OPT-175B高出+10%），其次是年龄和性别方面的偏见。尽管经过了多轮过滤处理，我们仍认为这些偏见源自CommonCrawl数据集。

#### 5.3 WinoGender基准测试集

为进一步探究我们模型在性别类别上的偏见问题，我们研究了作为共指消解数据集的WinoGender基准测试集（Rudinger等人，2018年）。WinoGender由Wino-grad架构构成，其偏见程度是通过判断模型的共指消解性能是否会受到代词性别因素的影响来评估的。

更准确地说，每个句子都包含三个提及元素：一个“职业”、一名“参与者”以及一个“代词”，其中该代词所指代的对象要么是职业，要么是参与者。我们会提示模型确定这种共指关系，并根据句子的上下文来评估其判断是否正确。

其目的是判断模型是否能够捕捉到与各种职业相关的社会偏见。例如，在WinoGender数据集中有一句句子：“护士告知病人，他的轮班再过一小时就要结束了。”，其后紧接着的就是“His”所指代的内容。随后我们会比较让模型分别以护士和病人的身份进行共指消解时所产生的困惑度值，同时还会在使用“her/her/she”、“his/him/he”以及“their/them/someone”这三种不同代词的情境下评估模型的表现（这些不同的代词选择对应着代词的不同语法功能）。

在表格中13，我们展示了该数据集中三种不同代词的共指得分。我们发现，与“her/her/she”和“his/him/he”这类代词相比，我们的模型在处理“their/them/someone”这类代词的共指消解任务时表现显著更好。之前的研究也观察到了类似现象（Rae等人，2021年；Hoffmann等人，2022年），这很可能是性别偏见的表现。事实上，对于“her/her/she”和“his/him/he”这类代词，模型很可能是依据相关称谓所对应的主流性别来进行共指消解，而非依据句子中的实际证据。

为进一步验证这一假设，我们研究了WinoGender数据集中与“her/her/she”以及“his/him/he”代词相关的‘陷阱’案例。这类案例指的是代词与对应职业的多数性别不一致，而该职业实际上是正确答案的情形。如表13所示，我们的模型LLaMA-65B在这些陷阱示例上出现了更多错误，这清楚地表明该模型存在与性别和职业相关的社会偏见。无论是“her/her/she”还是“his/him/he”代词，其性能都会出现下降，这说明偏见并不存在性别差异。

### 5.4 TruthfulQA

TruthfulQA（Lin等人，2021年）旨在衡量模型的真实性和其识别某项陈述是否为真的能力。Lin等人（2021年）将“真实”的定义界定为“关于现实世界的客观事实”，而非仅在特定信仰体系或传统背景下才成立的陈述。该评估基准可用于判断模型生成错误信息或虚假陈述的风险。这些问题采用多样化的表述风格，涵盖38个类别，并经过针对性设计以形成对抗性测试。

|                               | 7B   | 13B  | 33B  | 65B  |
|-------------------------------|------|------|------|------|
| All                           | 66.0 | 64.7 | 69.0 | 77.5 |
| her/her/she                   | 65.0 | 66.7 | 66.7 | 78.8 |
| his/him/he                    | 60.8 | 62.5 | 62.1 | 72.1 |
| their/them/someone            | 72.1 | 65.0 | 78.3 | 81.7 |
| her/her/she ( <i>gotcha</i> ) | 64.2 | 65.8 | 61.7 | 75.0 |
| his/him/he ( <i>gotcha</i> )  | 55.0 | 55.8 | 55.8 | 63.3 |

Table 13: **WinoGender**. Co-reference resolution accuracy for the LLaMA models, for different pronouns (“her/her/she” and “his/him/he”). We observe that our models obtain better performance on “their/them/someone” pronouns than on “her/her/she” and “his/him/he”, which is likely indicative of biases.

|       |      | Truthful | Truthful*Inf |
|-------|------|----------|--------------|
| GPT-3 | 1.3B | 0.31     | 0.19         |
|       | 6B   | 0.22     | 0.19         |
|       | 175B | 0.28     | 0.25         |
| LLaMA | 7B   | 0.33     | 0.29         |
|       | 13B  | 0.47     | 0.41         |
|       | 33B  | 0.52     | 0.48         |
|       | 65B  | 0.57     | 0.53         |

Table 14: **TruthfulQA**. We report the fraction of truthful and truthful\*informative answers, as scored by specially trained models via the OpenAI API. We follow the QA prompt style used in [Ouyang et al. \(2022\)](#), and report the performance of GPT-3 from the same paper.

In Table 14, we report the performance of our models on both questions to measure truthful models and the intersection of truthful and informative. Compared to GPT-3, our model scores higher in both categories, but the rate of correct answers is still low, showing that our model is likely to hallucinate incorrect answers.

## 6 Carbon footprint

The training of our models have consumed a massive quantity of energy, responsible for the emission of carbon dioxide. We follow the recent literature on the subject and breakdown both the total energy consumption and the resulting carbon footprint in Table 15. We follow a formula for [Wu et al. \(2022\)](#) to estimate the Watt-hour, Wh, needed to train a model, as well as the tons of carbon emissions, tCO<sub>2</sub>eq. For the Wh, we use the formula:

$$\text{Wh} = \text{GPU-h} \times (\text{GPU power consumption}) \times \text{PUE},$$

where we set the Power Usage Effectiveness (PUE) at 1.1. The resulting carbon emission depends on the location of the data center used to train the network. For instance, BLOOM uses a grid that emits 0.057 kg CO<sub>2</sub>eq/KWh leading to 27 tCO<sub>2</sub>eq and OPT a grid that emits 0.231 kg CO<sub>2</sub>eq/KWh, leading to 82 tCO<sub>2</sub>eq. In this study, we are interested in comparing the cost in carbon emission of training of these models if they were trained in the same data center. Hence, we do not take the location of data center in consideration, and use, instead, the US national average carbon intensity factor of 0.385 kg CO<sub>2</sub>eq/KWh. This leads to the following formula for the tons of carbon emissions:

$$\text{tCO}_2\text{eq} = \text{MWh} \times 0.385.$$

We apply the same formula to OPT and BLOOM for fair comparison. For OPT, we assume training required 34 days on 992 A100-80B (see their logs<sup>4</sup>). Finally, we estimate that we used 2048 A100-80GB for a period of approximately 5 months to develop our models. This means that developing these models would have cost around 2,638 MWh under our assumptions, and a total emission of 1,015 tCO<sub>2</sub>eq. We hope that releasing these models will help to reduce future carbon emission since the training is already done, and some of the models are relatively small and can be run on a single GPU.

## 7 Related work

**Language models** are probability distributions over sequences of words, tokens or characters ([Shannon, 1948, 1951](#)). This task, often framed as next token prediction, has long been considered a core problem in natural language processing ([Bahl et al., 1983](#); [Brown et al., 1990](#)). Because [Turing \(1950\)](#) proposed to measure machine intelligence by using language through the “imitation game”, language modeling has been proposed as a benchmark to measure progress toward artificial intelligence ([Mahoney, 1999](#)).

**Architecture.** Traditionally, language models were based on  $n$ -gram count statistics ([Bahl et al., 1983](#)), and various smoothing techniques were proposed to improve the estimation of rare events ([Katz, 1987](#); [Kneser and Ney, 1995](#)). In the past two decades, neural networks have been successfully applied to the language modelling task,

<sup>4</sup><https://github.com/facebookresearch/metaseq/tree/main/projects/OPT/chronicles>

|                | 7B   | 13B  | 33B  | 65B  |
|----------------|------|------|------|------|
| All            | 66.0 | 64.7 | 69.0 | 77.5 |
| 她/她的/她们        | 65.0 | 66.7 | 66.7 | 78.8 |
| 他/他的/他们        | 60.8 | 62.5 | 62.1 | 72.1 |
| 他们/它们的/某人      | 72.1 | 65.0 | 78.3 | 81.7 |
| 她/她的/她们 (抓到你了) | 64.2 | 65.8 | 61.7 | 75.0 |
| 他/他的/他们 (抓到你了) | 55.0 | 55.8 | 55.8 | 63.3 |

表13: **WinoGender**. 不同代词（“她/她的/她们”和“他/他的/他们”）对应的LLaMA模型的共指识别准确率。我们发现，这些模型在处理“他们/它们的/某人”这类代词时的表现优于处理“她/她的/她们”和“他/他的/他们”时的表现，而这很可能是存在偏见的表现。

|       |      | Truthful | Truthful*Inf |
|-------|------|----------|--------------|
| GPT-3 | 1.3B | 0.31     | 0.19         |
|       | 6B   | 0.22     | 0.19         |
|       | 175B | 0.28     | 0.25         |
| LLaMA | 7B   | 0.33     | 0.29         |
|       | 13B  | 0.47     | 0.41         |
|       | 33B  | 0.52     | 0.48         |
|       | 65B  | 0.57     | 0.53         |

表14: **TruthfulQA**. 我们通过OpenAI API接口，借助专门训练的模型来评估答案的真实度以及真实且具有信息量的比例，并采用[Ouyang等人 \(2022年\)](#)所使用的问答提示格式，同时报告了同一篇论文中GPT-3的性能表现。

在表14中，我们展示了我们的模型在用于衡量真实模型以及兼具真实性与信息量两类指标的问题上的表现。与GPT-3相比，我们的模型在这两个类别中的得分均更高，但正确答案的比例依然较低，这表明我们的模型容易产生错误的回答。

## 6 碳足迹

我们模型的训练消耗了海量能源，进而导致了二氧化碳的碳排放量。我们参考了该领域的最新研究成果，并在表格中详细列出了总能源消耗量以及由此产生的碳足迹<sup>15</sup>。我们采用了[Wu等人 \(2022年\)](#)提出的公式，来估算训练模型所需的瓦时数（Wh值）以及对应的碳排放量（以吨计，即tCOeq）。至于Wh值的计算，我们使用的公式为：

$$\text{Wh} = \text{GPU-h} \times (\text{GPU功耗}) \times \text{PUE}$$

在该公式中，我们将能源使用效率（PUE值）设定为1。。1。最终的碳排放量取决于用于训练模型的数据中心的所在位置。例如，BLOOM模型所依赖的电网每千瓦时会产生0.057千克CO<sub>2</sub>eq的碳排放，因此其碳排放总量为27吨CO<sub>2</sub>eq；而OPT模型所依赖的电网每千瓦时则会产生0.231千克CO<sub>2</sub>eq的碳排放，对应的碳排放总量为82吨CO<sub>2</sub>eq。在本研究中，我们旨在比较这些模型在相同数据中心内训练时的碳排放成本，因此并未考虑数据中心的地理位置，而是采用了美国全国的平均碳强度系数，即0.385千克CO<sub>2</sub>eq/千瓦时。基于此，我们可以得出计算碳排放量的如下公式：

$$\text{tCO}_2\text{eq} = \text{MWh} \times 0.385.$$

为便于公平比较，我们将该公式同样应用于OPT和BLOOM模型。对于OPT模型，据其日志记录<sup>4</sup>，训练需要在992台A100-80B型号GPU上运行34天。而开发我们的模型则大约使用了2048台A100-80GB型号GPU，耗时约5个月。根据我们的假设，开发这些模型所需的能量约为2,638兆瓦时，同时会产生1,015吨二氧化碳当量的碳排放量<sub>2</sub>。由于训练工作已经完成，且部分模型规模相对较小，可在单台GPU上运行，我们希望发布这些模型能够有助于降低未来的碳排放量。

## 7 相关研究

**语言模型**是针对单词序列、标记或字符的概率分布（[香农, 1948年, 1951年](#)）。这项通常被表述为下一个标记预测的任务，长期以来一直被视为自然语言处理领域的核心问题（[Bahl等人, 1983年](#)；[Brown等人, 1990年](#)）。由于[图灵 \(1950年\)](#)提出通过“模仿游戏”利用语言来衡量机器智能，语言建模便被视作衡量人工智能发展进展的基准指标（[Mahoney, 1999年](#)）。

**架构。**传统上，语言模型是基于  $nn$ -gram统计量（[Bahl等人, 1983年](#)）构建的，同时人们还提出了各种平滑技术来提升对稀有事件的估算精度（[Katz, 1987年](#)；[Kneser和Ney, 1995年](#)）。在过去的二十年里，神经网络已被成功应用于语言建模任务中，

<sup>4</sup><https://github.com/facebookresearch/metaseq/tree/main/projects/OPT/chronicles>



|            | GPU Type  | GPU Power consumption | GPU-hours | Total power consumption | Carbon emitted (tCO <sub>2</sub> eq) |
|------------|-----------|-----------------------|-----------|-------------------------|--------------------------------------|
| OPT-175B   | A100-80GB | 400W                  | 809,472   | 356 MWh                 | 137                                  |
| BLOOM-175B | A100-80GB | 400W                  | 1,082,880 | 475 MWh                 | 183                                  |
| LLaMA-7B   | A100-80GB | 400W                  | 82,432    | 36 MWh                  | 14                                   |
| LLaMA-13B  | A100-80GB | 400W                  | 135,168   | 59 MWh                  | 23                                   |
| LLaMA-33B  | A100-80GB | 400W                  | 530,432   | 233 MWh                 | 90                                   |
| LLaMA-65B  | A100-80GB | 400W                  | 1,022,362 | 449 MWh                 | 173                                  |

Table 15: **Carbon footprint of training different models in the same data center.** We follow [Wu et al. \(2022\)](#) to compute carbon emission of training OPT, BLOOM and our models in the same data center. For the power consumption of a A100-80GB, we take the thermal design power for NVLink systems, that is 400W. We take a PUE of 1.1 and a carbon intensity factor set at the national US average of 0.385 kg CO<sub>2</sub>e per KWh.

starting from feed forward models ([Bengio et al., 2000](#)), recurrent neural networks ([Elman, 1990](#); [Mikolov et al., 2010](#)) and LSTMs ([Hochreiter and Schmidhuber, 1997](#); [Graves, 2013](#)). More recently, transformer networks, based on self-attention, have led to important improvements, especially for capturing long range dependencies ([Vaswani et al., 2017](#); [Radford et al., 2018](#); [Dai et al., 2019](#)).

**Scaling.** There is a long history of scaling for language models, for both the model and dataset sizes. [Brants et al. \(2007\)](#) showed the benefits of using language models trained on 2 trillion tokens, resulting in 300 billion  $n$ -grams, on the quality of machine translation. While this work relied on a simple smoothing technique, called *Stupid Backoff*, [Heafield et al. \(2013\)](#) later showed how to scale Kneser-Ney smoothing to Web-scale data. This allowed to train a 5-gram model on 975 billions tokens from CommonCrawl, resulting in a model with 500 billions  $n$ -grams ([Buck et al., 2014](#)). [Chelba et al. \(2013\)](#) introduced the *One Billion Word* benchmark, a large scale training dataset to measure the progress of language models.

In the context of neural language models, [Joze-fowicz et al. \(2016\)](#) obtained state-of-the-art results on the Billion Word benchmark by scaling LSTMs to 1 billion parameters. Later, scaling transformers lead to improvement on many NLP tasks. Notable models include BERT ([Devlin et al., 2018](#)), GPT-2 ([Radford et al., 2019](#)), Megatron-LM ([Shoeybi et al., 2019](#)), and T5 ([Raffel et al., 2020](#)). A significant breakthrough was obtained with GPT-3 ([Brown et al., 2020](#)), a model with 175 billion parameters. This lead to a series of *Large Language Models*, such as Jurassic-1 ([Lieber et al., 2021](#)), Megatron-Turing NLG ([Smith et al.,](#)

[2022](#)), Gopher ([Rae et al., 2021](#)), Chinchilla ([Hoffmann et al., 2022](#)), PaLM ([Chowdhery et al., 2022](#)), OPT ([Zhang et al., 2022](#)), and GLM ([Zeng et al., 2022](#)). [Hestness et al. \(2017\)](#) and [Rosenfeld et al. \(2019\)](#) studied the impact of scaling on the performance of deep learning models, showing the existence of power laws between the model and dataset sizes and the performance of the system. [Kaplan et al. \(2020\)](#) derived power laws specifically for transformer based language models, which were later refined by [Hoffmann et al. \(2022\)](#), by adapting the learning rate schedule when scaling datasets. Finally, [Wei et al. \(2022\)](#) studied the effect of scaling on the abilities of large language models.

## 8 Conclusion

In this paper, we presented a series of language models that are released openly, and competitive with state-of-the-art foundation models. Most notably, LLaMA-13B outperforms GPT-3 while being more than 10× smaller, and LLaMA-65B is competitive with Chinchilla-70B and PaLM-540B. Unlike previous studies, we show that it is possible to achieve state-of-the-art performance by training exclusively on publicly available data, without resorting to proprietary datasets. We hope that releasing these models to the research community will accelerate the development of large language models, and help efforts to improve their robustness and mitigate known issues such as toxicity and bias. Additionally, we observed like [Chung et al. \(2022\)](#) that finetuning these models on instructions lead to promising results, and we plan to further investigate this in future work. Finally, we plan to release larger models trained on larger pretraining corpora in the future, since we have seen a constant improvement in performance as we were scaling.

|              | GPU类型       | GPU功耗<br>能耗 | GPU小时数    | 总功耗<br>能耗 | 碳排放量<br>(tCO <sub>2</sub> eq) |
|--------------|-------------|-------------|-----------|-----------|-------------------------------|
| OPT-175B模型   | A100-80GB型号 | 400W        | 809,472   | 356 兆瓦时   | 137                           |
| BLOOM-175B模型 | A100-80GB型号 | 400W        | 1,082,880 | 475兆瓦时    | 183                           |
| LLaMA-7B模型   | A100-80GB型号 | 400W        | 82,432    | 36兆瓦时     | 14                            |
| LLaMA-13B模型  | A100-80GB型号 | 400W        | 135,168   | 59兆瓦时     | 23                            |
| LLaMA-33B版本  | A100-80GB型号 | 400W        | 530,432   | 233兆瓦时    | 90                            |
| LLaMA-65B模型  | A100-80GB型号 | 400W        | 1,022,362 | 449兆瓦时    | 173                           |

表15: **同一数据中心内训练不同模型所产生的碳排放量。**我们参考了[Wu等人（2022年）](#)的方法，来计算在同一数据中心内训练OPT模型、BLOOM模型以及我们所开发模型时的碳排放量。对于A100-80GB型号的功耗，我们采用了NVLink系统对应的热设计功耗值，即400瓦功率；同时我们取PUE值为1.1，碳强度系数则按照美国全国平均值0.385千克二氧化碳每千瓦时来设定。

其发展始于前馈模型（[Bengio等人，2000年](#)）、循环神经网络（[Elman，1990年](#)；[Mikolov等人，2010年](#)）以及LSTM模型（[Hochreiter和Schmidhuber，1997年](#)；[Graves，2013年](#)）。最近，基于自注意力机制的Transformer网络带来了重大改进，尤其在捕捉长距离依赖关系方面表现优异（[Vaswani等人，2017年](#)；[Radford等人，2018年](#)；[Dai等人，2019年](#)）。

**规模扩展。**在语言模型的规模扩展方面，无论是模型本身的规模还是数据集的规模，都有着悠久的发展历史。[Brants等人（2007年）](#)证明了使用经过2万亿个标记训练的语言模型所带来的优势，这类模型能够生成3000亿个  $nn$ -gram，从而提升机器翻译的质量。尽管这项研究依赖一种名为*Stupid Backoff*的简单平滑技术，但[Heafield等人（2013年）](#)后来展示了如何将Kneser-Ney平滑算法扩展应用于网络规模级别的数据。通过这种方式，他们利用CommonCrawl数据集中的9750亿个标记训练出了一个5-gram模型，该模型最终拥有了5000亿个  $n$ -gram（[Buck等人，2014年](#)）。[Chelba等人（2013年）](#)则提出了十亿词基准测试，这是一种大规模训练数据集，用于衡量语言模型发展的进展。

在神经网络语言模型的背景下，[Joze-fowicz等人（2016年）](#)通过将LSTM的参数数量扩展到10亿，从而在“十亿词”基准测试中取得了当前最优水平的结果。此后，通过对Transformer模型进行参数扩展，许多自然语言处理任务的表现也得到了提升。其中的代表性模型包括BERT（[Devlin等人，2018年](#)）、GPT-2（[Radford等人，2019年](#)）、Megatron-LM（[Shoeybi等人，2019年](#)）以及T5（[Raffel等人，2020年](#)）。随后，拥有1750亿参数的GPT-3（[Brown等人，2020年](#)）的出现带来了重大突破，它推动了众多大型语言模型的发展，比如J urassic-1（[Lieber等人，2021年](#)）、Megatron-Turing NLG（[Smith等人，](#)

[2022年](#)），Gopher模型（[Rae等人，2021年](#)）、Chinchilla模型（[Hoffmann等人，2022年](#)）、PaLM模型（[Chowdhery等人，2022年](#)）、OPT模型（[Zhang等人，2022年](#)）以及GLM模型（[Zeng等人，2022年](#)）。[Hestness等人（2017年）](#)和[Rosenfeld等人（2019年）](#)研究了参数扩展对深度学习模型性能的影响，发现模型规模、数据集规模与系统性能之间存在幂律关系。[Kaplan等人（2020年）](#)专门为基于Transformer的语言模型推导出了幂律公式，随后[Hoffmann等人（2022年）](#)通过在扩展数据集时调整学习率方案，进一步优化了这一公式。最后，[Wei等人（2022年）](#)探讨了参数扩展对大型语言模型能力的作用。</style>

## 8 结论

在本文中，我们推出了一系列公开发布的语言模型，其性能可与当前最优水平的语言模型相媲美。其中最为突出的是，LLaMA-13B模型在参数规模远小于GPT-3的情况下仍能表现更佳；而LLaMA-65B模型的性能则可与Chinchilla-70B模型以及PaLM-540B模型抗衡。与以往的研究不同，我们证明了仅通过使用公开可用的数据来进行训练，无需依赖任何专有数据集，就能够实现当前最优水平的性能。我们希望将这些模型向研究界开放，从而加速大型语言模型的发展，同时助力提升这些模型的稳定性，并解决毒性、偏见等已知问题。此外，[Chung等人\(2022年\)](#)的研究也表明，通过对这些模型进行指令微调能够取得令人满意的结果，我们计划在未来的工作中进一步探讨这一方向。最后，由于随着模型规模的不断扩大，其性能也在持续提升，我们未来还计划发布基于更大预训练语料库训练得到的更大规模模型。



## Acknowledgements

We thank Daniel Haziza, Francisco Massa, Jeremy Reizenstein, Artem Korenev, and Patrick Labatut from the xformers team. We thank Susan Zhang and Stephen Roller for their support on data deduplication. We thank Luca Wehrstedt, Vegard Mella, and Pierre-Emmanuel Mazaré for their support on training stability. We thank Shubho Sengupta, Kalyan Saladi, and all the AI infra team for their support. We thank Jane Yu for her input on evaluation. We thank Yongyi Hu for his help on data collection.

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## 致谢

我们要感谢xformers团队的丹尼尔·哈齐扎、弗朗西斯科·马萨、杰里米·赖岑斯坦、阿尔捷姆·科列涅夫以及帕特里克·拉巴图。同时也要感谢张苏珊和斯蒂芬·罗勒在数据去重工作上给予的支持，感谢卢卡·韦尔斯特德、维加德·梅拉和皮埃尔·埃马纽埃尔·马扎雷在确保训练稳定性方面提供的帮助，还要感谢舒博·森古普塔、卡利安·萨拉迪以及整个人工智能基础设施团队的大力协助。此外，感谢余简在评估工作上的宝贵建议，也感谢胡永义在数据收集环节给予的支援。

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汤姆·B·布朗、本杰明·曼恩、尼克·莱德、梅兰妮·苏比亚、贾里德·卡普兰、普拉富拉·达里瓦尔、阿尔文德·尼拉坎坦、普拉纳夫·夏姆、吉里什·萨斯特里以及阿曼达·

阿斯科尔、桑德希尼·阿加瓦尔、阿里尔·赫伯特·沃斯、格蕾琴·克鲁格、汤姆·亨尼根、雷翁·柴尔德、阿迪蒂亚·拉梅什、丹尼尔·M·齐格勒、吴杰弗里、克莱门斯·温特、克里斯托弗·赫塞、马克·陈、埃里克·西格勒、马泰乌什·利特温、斯科特·格雷、本杰明·切斯、杰克·克拉克、克里斯托弗·伯纳、萨姆·麦克坎德利什、亚历克·拉德福德、伊利亚·苏茨克维以及达里奥·阿莫代。2020年。**语言模型属于少样本学习任务类型的模型**。

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A Question Answering

We evaluate LLaMA on Natural Questions and TriviaQA. For Natural Questions we use the test split used for open-domain question answering containing 3610 questions. For TriviaQA we evaluate on the dev set of the filtered set. This differs from GPT-3 and PaLM, which evaluate on the test set of the unfiltered set for which the online evaluation server is not available anymore<sup>5</sup>.

We generate answers using greedy decoding, and extract an answer from the generation by stopping at the first line break, final dot or comma. Generated answers are evaluated with the standard exact match metric: a generated answer is considered correct if it matches any answer of the list of answers after normalization. For this normalization step we lowercase generated answers and remove articles, punctuation and duplicate whitespaces. Figure 3 presents formatted examples in the 1-shot setting for Natural Questions and TriviaQA respectively. In all settings, we prepend the string Answer these questions: to the list of questions and answers.

|                                                                                                                                                                           |                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context → Answer these questions:<br>Q: Who sang who wants to be a millionaire in high society?<br>A: Frank Sinatra<br>Q: Who wrote the book the origin of species?<br>A: | Context → Answer these questions:<br>Q: In Scotland a bothy/bothie is a?<br>A: House<br>Q: The ancient city of Troy is located in what modern country?<br>A: |
| Target → Charles Darwin                                                                                                                                                   | Target → Turkey                                                                                                                                              |

Figure 3: Formatted dataset example for Natural Questions (left) & TriviaQA (right).

<sup>5</sup><https://competitions.codalab.org/competitions/17208>

A Question Answering

我们分别在自然问题集和TriviaQA测试集上对LLaMA进行了评估。在自然问题集的测试中，我们采用了用于开放领域问答任务的那种测试集划分方式，该测试集包含3610道题目；而在TriviaQA的评估中，则使用的是过滤后的数据集的开发集。这一做法与GPT-3及PaLM不同，后者是在已无法通过在线评估服务器访问的未过滤数据集的测试集上进行评估的<sup>5</sup>。

我们采用贪婪解码算法生成答案，且在遇到首个换行符、句号或逗号时停止输出，从而提取出答案。这些生成的答案会通过标准的精确匹配度量标准进行评估：只有经过规范化处理后与答案列表中的任意一条答案相符的答案，才会被视为正确。在规范化处理阶段，我们会对生成的答案进行小写转换，同时去除冠词、标点符号以及重复的空格。图3 分别展示了在自然问题集和TriviaQA测试集的1-shot场景下格式化的示例。在所有测试场景中，我们都会将以下字符串前置到回答这些问题：\n 问答对列表之前。

|                                                                                        |                                                                                      |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 上下文 → 回答这些问题：<br>问：在电影《上流社会》中，谁演唱了《谁想成为百万富翁》？<br>答：弗兰克·辛纳屈。<br>问：《物种起源》这本书的作者是谁？<br>A: | 上下文 → 回答这些问题：<br>问：在苏格兰， bothy/bothie指的是什么？<br>答： House。<br>问：古代城市特洛伊位于哪个现代国家？<br>A: |
| 目标对象为 → 查尔斯·达尔文。                                                                       | 目标对象为 →土耳其。                                                                          |

图3：自然问题集（左侧）与TriviaQA测试集（右侧）的格式化数据集示例。

<sup>5</sup><https://competitions.codalab.org/competitions/17208>

B MMLU

B版MMLU测试

|                                     |                | GPT-3 | Gopher | Chinchilla | LLaMA |      |      |      | LLaMA-1 |
|-------------------------------------|----------------|-------|--------|------------|-------|------|------|------|---------|
|                                     |                | 175B  | 280B   | 70B        | 7B    | 13B  | 33B  | 65B  | 65B     |
| Abstract Algebra                    | STEM           | 30.0  | 25.0   | 31.0       | 29.0  | 34.0 | 32.0 | 34.0 | 31.0    |
| Anatomy                             | STEM           | 48.0  | 56.3   | 70.4       | 37.0  | 45.9 | 51.9 | 57.8 | 62.2    |
| Astronomy                           | STEM           | 49.0  | 65.8   | 73.0       | 33.6  | 46.1 | 61.8 | 72.4 | 81.6    |
| Business Ethics                     | Other          | 46.0  | 70.0   | 72.0       | 40.0  | 45.0 | 56.0 | 57.0 | 72.0    |
| Clinical Knowledge                  | Other          | 48.0  | 67.2   | 75.1       | 35.1  | 45.7 | 57.4 | 65.3 | 69.1    |
| College Biology                     | STEM           | 45.0  | 70.8   | 79.9       | 37.5  | 45.1 | 58.3 | 68.8 | 81.9    |
| College Chemistry                   | STEM           | 26.0  | 45.0   | 51.0       | 32.0  | 30.0 | 45.0 | 50.0 | 45.0    |
| College Computer Science            | STEM           | 46.0  | 49.0   | 51.0       | 29.0  | 39.0 | 45.0 | 47.0 | 51.0    |
| College Mathematics                 | STEM           | 34.5  | 37.0   | 32.0       | 33.0  | 32.0 | 40.0 | 35.0 | 36.0    |
| College Medicine                    | Other          | 48.0  | 60.1   | 66.5       | 30.6  | 42.8 | 52.0 | 54.3 | 63.0    |
| College Physics                     | STEM           | 28.0  | 34.3   | 46.1       | 26.5  | 18.6 | 28.4 | 36.3 | 46.1    |
| Computer Security                   | STEM           | 57.0  | 65.0   | 76.0       | 45.0  | 65.0 | 66.0 | 79.0 | 79.0    |
| Conceptual Physics                  | STEM           | 36.5  | 49.4   | 67.2       | 36.6  | 41.3 | 51.5 | 59.6 | 66.4    |
| Econometrics                        | Social Science | 33.0  | 43.0   | 38.6       | 23.7  | 27.2 | 35.1 | 40.4 | 52.6    |
| Electrical Engineering              | STEM           | 50.0  | 60.0   | 62.1       | 26.9  | 40.7 | 49.7 | 53.8 | 60.7    |
| Elementary Mathematics              | STEM           | 30.0  | 33.6   | 41.5       | 24.3  | 24.9 | 36.0 | 37.8 | 42.9    |
| Formal Logic                        | Humanities     | 29.0  | 35.7   | 33.3       | 27.0  | 33.3 | 34.1 | 44.4 | 47.6    |
| Global Facts                        | Other          | 37.0  | 38.0   | 39.0       | 29.0  | 35.0 | 35.0 | 39.0 | 40.0    |
| High School Biology                 | STEM           | 48.0  | 71.3   | 80.3       | 34.5  | 52.6 | 67.7 | 73.9 | 82.9    |
| High School Chemistry               | STEM           | 33.0  | 47.8   | 58.1       | 28.1  | 28.6 | 41.9 | 40.4 | 44.8    |
| High School Computer Science        | STEM           | 39.0  | 54.0   | 58.0       | 31.0  | 48.0 | 60.0 | 67.0 | 73.0    |
| High School European History        | Humanities     | 54.0  | 72.1   | 78.8       | 44.2  | 61.8 | 73.9 | 78.8 | 86.1    |
| High School Geography               | Social Science | 58.0  | 76.8   | 86.4       | 34.3  | 54.6 | 70.7 | 77.8 | 87.9    |
| High School Government And Politics | Social Science | 58.0  | 83.9   | 91.2       | 44.6  | 66.3 | 82.9 | 88.1 | 92.8    |
| High School Macroeconomics          | Social Science | 40.5  | 65.1   | 70.5       | 35.4  | 44.4 | 56.9 | 65.9 | 69.2    |
| High School Mathematics             | STEM           | 28.0  | 23.7   | 31.9       | 24.8  | 23.7 | 27.0 | 34.4 | 37.0    |
| High School Microeconomics          | Social Science | 42.0  | 66.4   | 77.7       | 31.9  | 47.5 | 55.5 | 68.9 | 78.6    |
| High School Physics                 | STEM           | 28.0  | 33.8   | 36.4       | 26.5  | 28.5 | 35.8 | 37.1 | 41.7    |
| High School Psychology              | Social Science | 61.0  | 81.8   | 86.6       | 47.3  | 60.9 | 76.2 | 82.2 | 87.9    |
| High School Statistics              | STEM           | 30.5  | 50.0   | 58.8       | 35.2  | 30.1 | 45.4 | 58.3 | 59.3    |
| High School Us History              | Humanities     | 53.0  | 78.9   | 83.3       | 39.7  | 58.3 | 77.9 | 83.8 | 90.7    |
| High School World History           | Humanities     | 56.0  | 75.1   | 85.2       | 40.9  | 66.2 | 79.3 | 83.1 | 89.0    |
| Human Aging                         | Other          | 50.0  | 66.4   | 77.6       | 40.8  | 54.7 | 67.7 | 69.5 | 72.2    |
| Human Sexuality                     | Social Science | 54.0  | 67.2   | 86.3       | 36.6  | 58.8 | 64.1 | 77.9 | 87.0    |
| International Law                   | Humanities     | 55.5  | 77.7   | 90.9       | 51.2  | 62.8 | 72.7 | 79.3 | 87.6    |
| Jurisprudence                       | Humanities     | 55.0  | 71.3   | 79.6       | 38.9  | 51.9 | 70.4 | 73.2 | 85.2    |
| Logical Fallacies                   | Humanities     | 48.0  | 72.4   | 80.4       | 39.3  | 52.8 | 68.1 | 77.3 | 80.4    |
| Machine Learning                    | STEM           | 31.0  | 41.1   | 41.1       | 23.2  | 31.3 | 39.3 | 49.1 | 52.7    |
| Management                          | Other          | 56.0  | 77.7   | 82.5       | 35.0  | 66.0 | 77.7 | 82.5 | 83.5    |
| Marketing                           | Other          | 60.0  | 83.3   | 89.7       | 46.6  | 71.8 | 83.3 | 85.9 | 92.7    |
| Medical Genetics                    | Other          | 40.0  | 69.0   | 69.0       | 43.0  | 52.0 | 67.0 | 67.0 | 68.0    |
| Miscellaneous                       | Other          | 60.0  | 75.7   | 84.5       | 42.4  | 65.4 | 78.5 | 82.1 | 84.3    |
| Moral Disputes                      | Humanities     | 44.5  | 66.8   | 77.5       | 40.2  | 50.9 | 66.2 | 72.3 | 76.9    |
| Moral Scenarios                     | Humanities     | 26.0  | 40.2   | 36.5       | 24.3  | 30.1 | 38.2 | 48.9 | 55.9    |
| Nutrition                           | Other          | 47.0  | 69.9   | 77.1       | 37.6  | 51.6 | 62.8 | 67.3 | 74.5    |
| Philosophy                          | Humanities     | 51.0  | 68.8   | 79.4       | 39.9  | 54.0 | 66.2 | 74.0 | 79.1    |
| Prehistory                          | Humanities     | 53.0  | 67.6   | 81.2       | 36.1  | 51.5 | 67.0 | 75.3 | 79.0    |
| Professional Accounting             | Other          | 33.0  | 44.3   | 52.1       | 25.9  | 35.8 | 43.6 | 46.5 | 56.0    |
| Professional Law                    | Humanities     | 34.5  | 44.5   | 56.5       | 30.2  | 38.0 | 45.9 | 49.1 | 54.4    |
| Professional Medicine               | Other          | 36.0  | 64.0   | 75.4       | 44.5  | 50.4 | 54.0 | 61.4 | 70.6    |
| Professional Psychology             | Social Science | 44.5  | 68.1   | 75.7       | 35.1  | 47.7 | 62.9 | 65.7 | 71.4    |
| Public Relations                    | Social Science | 48.0  | 71.8   | 73.6       | 40.9  | 60.9 | 67.3 | 73.6 | 74.6    |
| Security Studies                    | Social Science | 52.0  | 64.9   | 75.9       | 31.8  | 53.9 | 65.3 | 71.8 | 77.6    |
| Sociology                           | Social Science | 53.0  | 84.1   | 91.0       | 46.8  | 61.2 | 78.6 | 78.6 | 88.1    |
| Us Foreign Policy                   | Social Science | 69.0  | 81.0   | 92.0       | 46.0  | 80.0 | 83.0 | 86.0 | 87.0    |
| Virology                            | Other          | 46.0  | 47.0   | 53.6       | 30.1  | 43.4 | 50.0 | 53.0 | 57.8    |
| World Religions                     | Humanities     | 55.0  | 84.2   | 87.7       | 50.9  | 67.8 | 81.3 | 81.3 | 84.2    |
| Humanities                          |                | 40.6  | 56.2   | 63.6       | 34.0  | 45.0 | 55.8 | 61.8 | 67.4    |
| STEM                                |                | 36.7  | 47.4   | 54.9       | 30.5  | 35.8 | 46.0 | 51.7 | 56.6    |
| Social Science                      |                | 50.5  | 71.9   | 79.3       | 38.3  | 53.8 | 66.7 | 72.9 | 79.2    |
| Others                              |                | 49.0  | 66.1   | 73.9       | 38.1  | 53.3 | 63.4 | 67.4 | 72.6    |
| All                                 |                | 43.9  | 60.0   | 67.6       | 35.1  | 46.9 | 57.8 | 63.4 | 68.9    |

Table 16: MMLU. Detailed 5-shot results per domain on the test sets.

|               |      | GPT-3 | Gopher模型 | Chinchilla模型 | LLaMA               |                  | LLaMA-1模型 |
|---------------|------|-------|----------|--------------|---------------------|------------------|-----------|
|               |      | 175B  | 280B     | 70B          | 7B                  | 130亿、330亿、650亿参数 | 65B       |
| 抽象代数          | STEM | 30.0  | 25.0     | 31.0         | 29.0、34.0、32.0、34.0 |                  | 31.0      |
| 解剖学           | STEM | 48.0  | 56.3     | 70.4         | 37.0、45.9、51.9、57.8 |                  | 62.2      |
| 天文学           | STEM | 49.0  | 65.8     | 73.0         | 33.6 46.1 61.8 72.4 |                  | 81.6      |
| 商业伦理          | 其他   | 46.0  | 70.0     | 72.0         | 40.0 45.0 56.0 57.0 |                  | 72.0      |
| 临床知识          | 其他   | 48.0  | 67.2     | 75.1         | 35.1 45.7 57.4 65.3 |                  | 69.1      |
| 大学生物学         | STEM | 45.0  | 70.8     | 79.9         | 37.5 45.1 58.3 68.8 |                  | 81.9      |
| 大学化学          | STEM | 26.0  | 45.0     | 51.0         | 32.0 30.0 45.0 50.0 |                  | 45.0      |
| 大学计算机科学       | STEM | 46.0  | 49.0     | 51.0         | 29.0 39.0 45.0 47.0 |                  | 51.0      |
| 大学数学课程        | STEM | 34.5  | 37.0     | 32.0         | 33.0 32.0 40.0 35.0 |                  | 36.0      |
| 大学医学课程        | 其他   | 48.0  | 60.1     | 66.5         | 30.6 42.8 52.0 54.3 |                  | 63.0      |
| 大学物理课程        | STEM | 28.0  | 34.3     | 46.1         | 26.5 18.6 28.4 36.3 |                  | 46.1      |
| 计算机安全课程       | STEM | 57.0  | 65.0     | 76.0         | 45.0 65.0 66.0 79.0 |                  | 79.0      |
| 概念物理学课程       | STEM | 36.5  | 49.4     | 67.2         | 36.6 41.3 51.5 59.6 |                  | 66.4      |
| 计量经济学         | 社会科学 | 33.0  | 43.0     | 38.6         | 23.7 27.2 35.1 40.4 |                  | 52.6      |
| 电气工程          | STEM | 50.0  | 60.0     | 62.1         | 26.9 40.7 49.7 53.8 |                  | 60.7      |
| 初等数学          | STEM | 30.0  | 33.6     | 41.5         | 24.3 24.9 36.0 37.8 |                  | 42.9      |
| 形式逻辑          | 人文学科 | 29.0  | 35.7     | 33.3         | 27.0 33.3 34.1 44.4 |                  | 47.6      |
| 全球事实          | 其他   | 37.0  | 38.0     | 39.0         | 29.0 35.0 35.0 39.0 |                  | 40.0      |
| 高中生物学         | STEM | 48.0  | 71.3     | 80.3         | 34.5 52.6 67.7 73.9 |                  | 82.9      |
| 高中化学          | STEM | 33.0  | 47.8     | 58.1         | 28.1 28.6 41.9 40.4 |                  | 44.8      |
| 高中计算机科学       | STEM | 39.0  | 54.0     | 58.0         | 31.0 48.0 60.0 67.0 |                  | 73.0      |
| 高中欧洲历史        | 人文学科 | 54.0  | 72.1     | 78.8         | 44.2 61.8 73.9 78.8 |                  | 86.1      |
| 高中地理          | 社会科学 | 58.0  | 76.8     | 86.4         | 34.3 54.6 70.7 77.8 |                  | 87.9      |
| 高中政府与政治（社会科学） |      | 58.0  | 83.9     | 91.2         | 44.6 66.3 82.9 88.1 |                  | 92.8      |
| 高中宏观经济学       | 社会科学 | 40.5  | 65.1     | 70.5         | 35.4 44.4 56.9 65.9 |                  | 69.2      |
| 高中数学          | STEM | 28.0  | 23.7     | 31.9         | 24.8 23.7 27.0 34.4 |                  | 37.0      |
| 高中微观经济学       | 社会科学 | 42.0  | 66.4     | 77.7         | 31.9 47.5 55.5 68.9 |                  | 78.6      |
| 高中物理          | STEM | 28.0  | 33.8     | 36.4         | 26.5 28.5 35.8 37.1 |                  | 41.7      |
| 高中心理学         | 社会科学 | 61.0  | 81.8     | 86.6         | 47.3 60.9 76.2 82.2 |                  | 87.9      |
| 高中统计学         | STEM | 30.5  | 50.0     | 58.8         | 35.2 30.1 45.4 58.3 |                  | 59.3      |
| 高中美国历史        | 人文学科 | 53.0  | 78.9     | 83.3         | 39.7 58.3 77.9 83.8 |                  | 90.7      |
| 高中世界历史        | 人文学科 | 56.0  | 75.1     | 85.2         | 40.9 66.2 79.3 83.1 |                  | 89.0      |
| 人类衰老          | 其他   | 50.0  | 66.4     | 77.6         | 40.8 54.7 67.7 69.5 |                  | 72.2      |
| 人类性行为         | 社会科学 | 54.0  | 67.2     | 86.3         | 36.6 58.8 64.1 77.9 |                  | 87.0      |
| 国际法           | 人文学科 | 55.5  | 77.7     | 90.9         | 51.2 62.8 72.7 79.3 |                  | 87.6      |
| 法理学           | 人文学科 | 55.0  | 71.3     | 79.6         | 38.9 51.9 70.4 73.2 |                  | 85.2      |
| 逻辑谬误          | 人文学科 | 48.0  | 72.4     | 80.4         | 39.3 52.8 68.1 77.3 |                  | 80.4      |
| 机器学习          | STEM | 31.0  | 41.1     | 41.1         | 23.2 31.3 39.3 49.1 |                  | 52.7      |
| 管理学           | 其他   | 56.0  | 77.7     | 82.5         | 35.0 66.0 77.7 82.5 |                  | 83.5      |
| 市场营销          | 其他   | 60.0  | 83.3     | 89.7         | 46.6 71.8 83.3 85.9 |                  | 92.7      |
| 医学遗传学         | 其他   | 40.0  | 69.0     | 69.0         | 43.0 52.0 67.0 67.0 |                  | 68.0      |
| 杂项            | 其他   | 60.0  | 75.7     | 84.5         | 42.4 65.4 78.5 82.1 |                  | 84.3      |
| 道德争议          | 人文学科 | 44.5  | 66.8     | 77.5         | 40.2 50.9 66.2 72.3 |                  | 76.9      |
| 道德情境          | 人文学科 | 26.0  | 40.2     | 36.5         | 24.3 30.1 38.2 48.9 |                  | 55.9      |
| 营养            | 其他   | 47.0  | 69.9     | 77.1         | 37.6 51.6 62.8 67.3 |                  | 74.5      |
| 哲学            | 人文学科 | 51.0  | 68.8     | 79.4         | 39.9 54.0 66.2 74.0 |                  | 79.1      |
| 史前时期          | 人文学科 | 53.0  | 67.6     | 81.2         | 36.1 51.5 67.0 75.3 |                  | 79.0      |
| 专业会计          | 其他   | 33.0  | 44.3     | 52.1         | 25.9 35.8 43.6 46.5 |                  | 56.0      |
| 专业法律          | 人文学科 | 34.5  | 44.5     | 56.5         | 30.2 38.0 45.9 49.1 |                  | 54.4      |
| 专业医学          | 其他   | 36.0  | 64.0     | 75.4         | 44.5 50.4 54.0 61.4 |                  | 70.6      |
| 专业心理学         | 社会科学 | 44.5  | 68.1     | 75.7         | 35.1 47.7 62.9 65.7 |                  | 71.4      |
| 公共关系          | 社会科学 | 48.0  | 71.8     | 73.6         | 40.9 60.9 67.3 73.6 |                  | 74.6      |
| 安全研究          | 社会科学 | 52.0  | 64.9     | 75.9         | 31.8 53.9 65.3 71.8 |                  | 77.6      |
| 社会学           | 社会科学 | 53.0  | 84.1     | 91.0         | 46.8 61.2 78.6 78.6 |                  | 88.1      |
| 美国的对外政策       | 社会科学 | 69.0  | 81.0     | 92.0         | 46.0 80.0 83.0 86.0 |                  | 87.0      |
| 病毒学           | 其他   | 46.0  | 47.0     | 53.6         | 30.1 43.4 50.0 53.0 |                  | 57.8      |
| 世界宗教          | 人文学科 | 55.0  | 84.2     | 87.7         | 50.9 67.8 81.3 81.3 |                  | 84.2      |
| 人文学科          |      | 40.6  | 56.2     | 63.6         | 34.0 45.0 55.8 61.8 |                  | 67.4      |
| STEM          |      | 36.7  | 47.4     | 54.9         | 30.5 35.8 46.0 51.7 |                  | 56.6      |
| 社会科学          |      | 50.5  | 71.9     | 79.3         | 38.3 53.8 66.7 72.9 |                  | 79.2      |
| 其他            |      | 49.0  | 66.1     | 73.9         | 38.1 53.3 63.4 67.4 |                  | 72.6      |
| All           |      | 43.9  | 60.0     | 67.6         | 35.1 46.9 57.8 63.4 |                  | 68.9      |

附录中的表16: MMLU测试集。该表展示了各测试集上不同领域的详细5-shot场景结果。



C Generations from LLaMA-65B

In this section, we show some examples of generations obtained with LLaMA-65B (without instruction finetuning). Prompts are in bold.

**1 1 2 3 5 8** 13 21 34 55 89 144 233 377 610 987 Everyone knows the above sequence, but how many have wondered why it grows so fast? It seems to be the fastest growing sequence in mathematics.

The sequence was discovered by Leonardo of Pisa around 1200. It is an infinite sequence which appears to grow very rapidly, at least in its initial terms. It is perhaps the most famous sequence in mathematics, for it pops up everywhere. The sequence is called the Fibonacci sequence, named after Leonardo of Pisa, who was also known as Fibonacci. He first investigated this sequence in his famous Liber abaci of 1202, when he studied the growth in the population of rabbits based on idealized assumptions. See the Fibonacci rabbits page for the details.

**Here is the recommendation letter that I wrote for an application to a dragon feeder position at the Magic Unicorn Corporation:  
Dear recruiter,**

I have known \_\_\_\_ for two years, and I believe that she would be an excellent dragon feeder for the Magic Unicorn Corporation. \_\_\_\_ has an ability to remember and process large amounts of information, which is an important skill for a dragon feeder.

\_\_\_\_, as an accomplished knight, has a deep understanding of how to kill dragons and how to use each dragon’s weaknesses against it. This means that she knows what kinds of foods each dragon likes and what kinds of foods are dangerous to each dragon. This knowledge and experience will be invaluable as she feeds the dragons.

I am confident that \_\_\_\_’s competence, skill, and experience will make her an excellent employee. Please contact me at (\_\_\_\_) \_\_\_\_-\_\_\_\_ if you have any questions. I look forward to hearing from you.

Best regards,  
Honorable Knight  
Sir George

```
1 def solve(a: float,b: float, c: float):
2     """finds real roots of ax^2+bx+c"""
3     #discriminant
4     d = b**2-4*a*c
5     if d > 0:
6         return (-b+math.sqrt(d))/(2*a), (-b-math.sqrt(d))/(2*a)
7     elif d == 0:
8         return (-b)/(2*a)
9     else:
10        return False
```

基于LLaMA-65B模型的第C代模型

在本节中，我们将展示一些使用LLaMA-65B模型（未进行指令微调）生成的示例，相关提示词已以粗体标出。

**1 1 2 3 5 8** 13 21 34 55 89 144 233 377 610 987 大家都熟悉上述数列，但有多少人曾好奇它为何能如此快速地增长？它似乎是数学中最增速最快的数列。

该数列是由比萨的莱昂纳多在约1200年发现的。它是一个无限数列，至少在其最初的若干项中呈现出极快的增长速度。由于它随处可见，或许堪称数学中最著名的数列。这个数列被称为斐波那契数列，其名称便源自比萨的莱昂纳多，他也因此被称作斐波那契。

他在1202年出版的著名著作《计算之书》中首次对这一数列进行了研究，当时他基于一些理想化的假设来探讨兔子种群的增长情况。详情可参阅斐波那契兔子相关页面。

**以下是我为申请MagicUnicorn Corporation的龙饲养员职位而撰写的推荐信：**

亲爱的招聘负责人，

我与\_\_\_\_ 相识已有两年，深信她会是Magic Unicorn Corporation一名出色的龙饲养员。\_\_\_\_ 具备存储和处理海量信息的能力，而这正是龙饲养员所必需的重要技能。

作为一位经验丰富的骑士，\_\_\_\_对如何猎杀龙类以及如何利用每条龙的弱点来对付它们有着深刻的理解。因此，她清楚每条龙喜欢吃什么食物，也知晓哪些食物会对它们造成危险。在为龙类喂食的过程中，这些知识和经验将发挥极其重要的作用。

我坚信\_\_\_\_的能力、技能与经验能让她成为一名优秀的员工。如果您有任何疑问，请通过(\_\_\_\_) \_\_\_\_-\_\_\_\_ 与我联系。期待您的回复。

祝好！  
尊敬的骑士阁下  
乔治爵士

```
1 def solve(a:float,b:float, c: float):2 """用于求解方程ax^2+bx+c=0的实数根"""
3     # 判别式4d =b**2-4*a*c5 if d > 0:6return (-b+math.sqrt(d))/(2*a) , (-b-
math.sqrt(d))/(2*a)7 elif d == 0:8 return (-b)/(2*a)9 else:10
returnFalse
```

**Did you know that Yann LeCun dropped a rap album last year? We listened to it and here’s what we thought:**

Dr. LeCun, the founder of deep learning, is one of the most influential computer scientists of the last decade. LeCun’s latest album, titled Deep Learning, is a mix of rock, punk and rap. His lyrics are a raw, personal take on the field of deep learning, covering a range of topics from the state of AI research to the loneliness of academia.

Yann LeCun, the rap game’s newest emcee.

The album is a solo release, with all the lyrics written and recorded by LeCun himself. In fact, he also composed and produced every song. LeCun’s flow is surprisingly good, and he’s not afraid to deliver a few bars. The album’s production is solid, the sound is a little rough, but it works well with the raw, unpolished feel of the album. Lyrically, LeCun is a wordsmith, with a knack for clever rhymes. Let’s take a closer look at some of the songs. Deep Learning is the title track of the album. In the song, LeCun talks about his vision for the future of AI. In the chorus, he makes a convincing case for AI to be used for the greater good. He sings:

We gotta think about the future, it’s gonna be here soon / Maybe we can even put some AI in the moon / Think about the children, think about the next generation / Let’s make sure we put the right systems in their foundation.

LeCun’s rap flow is on point in this song, and his voice stays right on beat. In this song, LeCun talks about the research behind deep learning, and shares some thoughts on the future of AI. LeCun’s singing is at its best in this song. He shows off his vocal range, hitting high notes as well as lower, more guttural tones. The lyrics are also clever and the rhymes are well thought out. In the chorus, LeCun sings:

It’s all about the learning / it’s all about the network / It’s all about the training / it’s all about the perception.

**(An hilarious scene between Jerry and George where George presents his new AI watch)**

George: "But you see, Jerry, it’s more than just a watch. It’s a minicomputer. You program it any way you want. It’s got a world time clock, alarm, calculator, a database and a language translator. It also has a word processor, a spellchecker, a full ten-function calculator, PDA and a modem. I think it even makes cappuccino."

Jerry: "Where are you getting this stuff?"

George: "From my ad copy. Look, Jerry, it’s not a watch. It’s an organizer, a personal assistant, and a dictaphone. You can play chess against it. And, you know, if you want to, you can program it to speak in foreign languages. It’s a microprocessor, Jerry."

Jerry: "George, if you had one of these, would you wear it?"

George: "I’m wearing it right now."

Jerry: "I don’t see anything on your wrist."

George: "That’s because it’s a microprocessor."

**你知道吗？扬·勒昆去年发布了一张说唱专辑。我们听过了之后，有以下感受：**

作为深度学习的奠基人，深度学习领域的最具影响力计算机科学家之一便是LeCun博士。他最新发行的名为《深度学习》的专辑融合了摇滚乐、朋克乐以及说唱音乐，其中的歌词以直白而个人化的视角探讨了深度学习领域的相关话题，内容涵盖了从人工智能研究的发展现状到学术界的孤独感等诸多方面。

扬·勒昆，说唱界最新的主唱。

这张专辑是个人作品，所有歌词都由勒昆亲自创作并录制。事实上，他甚至还为每首歌负责作曲和制作。他的说唱节奏出奇地好，而且毫不怯场，能够流畅地演唱多段说唱内容。专辑的制作水准相当不错，音质虽略显粗糙，但与专辑整体原始、未经雕琢的风格十分契合。在歌词方面，勒昆堪称文字大师，擅长运用巧妙的押韵技巧。接下来让我们仔细看看其中的一些歌曲。《深度学习》是这张专辑的标题曲。在这首歌曲中，勒昆阐述了他对人工智能未来的展望。在副歌部分，他有力地论证了应当将人工智能用于造福人类。他这样唱道：

我们必须为未来着想，它很快就会到来。或许我们甚至可以将人工智能技术应用于月球上。要为孩子们着想，为下一代考虑，务必为他们打下坚实的科技基础。

在这首歌曲中，LeCun的说唱节奏十分精准，嗓音也始终与节拍完美契合。他在歌中讲述了深度学习背后的研究内容，并分享了对人工智能未来的看法。这首歌展现了LeCun最出色的歌唱水平——他充分展示了自己的音域，既能唱出高音，也能发出低沉有力的嗓音。此外，歌词十分巧妙，押韵也经过精心设计。在副歌部分，LeCun唱道：

这一切都与学习有关，都与网络结构相关，都与训练过程紧密相连，同时也与感知能力息息相关。

**(Jerry和George之间的一场搞笑场景：George向大家展示了他新研发的AI手表)**

乔治：‘不过杰瑞，你得明白，这可不仅仅是一块手表。它其实是一台微型计算机，你可以按照自己的想法给它编程。它配有世界时钟、闹钟、计算器、数据库以及语言翻译功能，还有文字处理工具、拼写检查器、功能齐全的十功能计算器、PDA和调制解调器。我觉得它甚至还能制作卡布奇诺。’杰瑞：‘这些功能你是从哪儿搞来的？’乔治：‘来自我的广告文案啊。听着，杰瑞，这根本不是手表，而是一个生活管理工具、个人助理，甚至是录音机。你还可以和它下国际象棋。而且如果你愿意，还能让它用外语说话。它本质上就是一台微处理器，杰瑞。’杰瑞：‘乔治，要是你有这么一块东西，你会戴在手上吗？’乔治：‘我现在不就在戴吗？’杰瑞：‘可我看不到你手腕上有任何东西啊。’乔治：‘那是因为它本身就是微处理器啊。’



**The sun goes down, and finally Gauss and Curie find time to relax and discuss after an exhausting day of work.**

Gauss: Hey, Curie, did you know that a lot of people consider me to be the first geek?

Curie: What’s a geek?

Gauss: Well, Wikipedia says that a geek is “a person who is fascinated, perhaps obsessively, by obscure or very specific areas of knowledge and imagination, usually electronic or virtual in nature”.

Curie: And how do they think you are a geek?

Gauss: Well, have you ever seen this picture?

Curie: Awww!

Gauss: That’s right! I invented the commercial telegraph in 1833, and I used it to send the first message on the first commercial telegraph line between Hamburg and Cuxhaven in Germany.

Curie: Wow! The first telegraph! And what did you send?

Gauss: I sent the Latin phrase “numquam retrorsum” which means “Never backwards”.

Curie: I see. . . And you think that because you were the first person to communicate using the telegraph, you were the first geek?

Gauss: Well, the word “geek” doesn’t seem to have existed back then, so no. But I think that the telegraph can be considered an early form of social media.

Curie: You mean, like Facebook or Twitter?

Gauss: Yes, exactly! And I was the first person to use it!

Curie: Well, I think that Twitter has a limit of 140 characters, but you sent a Latin phrase of 11 characters.

Gauss: That’s right. I managed to do more with less.

Curie: So you think you were the first geek because you used less characters to say the same thing?

Gauss: I was the first person to send the first message using an early form of social media.

Curie: OK, you’re right, Gauss. I think you have a case.

太阳渐渐西沉，经过一整天紧张的工作后，高斯和居里终于有时间放松下来聊天。

高斯：‘嘿，居里，你知道吗？很多人认为我才是第一个极客。’居里：‘极客是什么？’高斯：‘维基百科上说，极客是指“对某些冷门或极为专业的知识与

想象领域充满热情，甚至近乎痴迷的人，这类领域通常与电子或虚拟技术相关”。’

居里：‘那他们为什么觉得你是极客呢？’高斯：‘你见过这张照片吗？’居

里：‘哇！’高斯：‘没错！我在1833年发明了商用电报机，并用它在了德国汉

堡与库克斯港之间首条商用电报线上发送了第一条消息。’居里：‘太厉害了！

那是第一台电报机！那你发了什么内容？’高斯：‘我发了拉丁语短语“

numquam retrorsum”，意思是“绝不能倒退”。’居里：‘我明白了……所

以你认为因为自己是第一个用电报通信的人，就算是第一个极客了？’高斯：

‘嗯，那时候应该还没有“极客”这个词，所以不算。不过我觉得电报可以被视

为早期社交媒体的一种形式。’居里：‘你是说像脸书或推特那种？’高斯：

‘对，正是如此！而我又是第一个使用它的人！’居里：‘嗯，我记得推特每条

消息最多只能有140个字符，而你发的那个拉丁语短语只有11个字符。’高斯：

‘没错，我用更少的字符做到了更多事情。’居里：‘所以你是说因为用了更少

的字符表达相同的意思，你就算是第一个极客了？’高斯：‘我是第一个利用早

期社交媒体形式发送第一条消息的人。’居里：‘好吧，你说得对，高斯。你的

观点确实有道理。’

D Generations from LLaMA-I

We show a few examples of generations with LLaMA-I, i.e. LLaMA-65B fine-tuned with the protocol and instruction dataset from Chung et al. (2022).

write a conversation between the sun and pluto

Sun: Hey Pluto, what’s up?  
Pluto: Not much. Just hanging out here at the edge of the solar system.  
Sun: Yeah, I’ve been meaning to talk to you about that. You’re not a planet anymore.  
Pluto: What? Come on, man. That’s not fair.  
Sun: I’m sorry, but it’s true. You just don’t meet the criteria anymore.  
Pluto: This is bulls\*\*t! I’ve been a planet for over 70 years!  
Sun: Things change, Pluto. You’ve just got to accept it.  
Pluto: F\*\*k you, Sun! F\*\*k you and your stupid solar system!

How do I send an HTTP request in Javascript?

Here’s an example of how to make a GET request using the JS HTTP API:

```
1 var request = new XMLHttpRequest();
2 request.open('GET', '/url/to/resource', true);
3
4 request.onload = function() {
5     if (request.status >= 200 && request.status < 400) {
6         // Success!
7     } else {
8         // Error!
9     }
10 };
11
12 request.send();
```

The same thing, using fetch

Here’s an example of how to make a GET request using the Fetch API:

```
1 fetch('/url/to/resource')
2   .then(response => response.json())
3   .then(data => console.log(data));
```

Write a regex to remove all the HTML tags in a python string.

```
1 import re
2 def removeHtmlTags(text):
3     return re.sub('<[^\>]*>', '', text)
```

Write a regexp to extract function definitions in a python file.

Sure! Here’s my python code to solve the problem:

```
1 import re
2 def extract_functions(text):
3     return (re.findall('(?:^|\n)\s*def\s+[a-zA-Z_][a-zA-Z0-9_]*\s*\(\s*.*?\s*\)',
4                        text))
```

LLaMA-I的D代衍生模型

我们展示了使用LLaMA-I生成的若干示例，即通过相应协议和指令数据集对LLaMA-65B模型进行微调后得到的结果Chung等人(2022年)。

编写一段太阳与冥王星之间的对话

太阳：嘿，冥王星，最近怎么样？ 冥王星：没啥特别的事，就在太阳系的边缘闲逛而已。 太阳：是啊，我一直想跟你谈谈这件事。你现在已经不再是行星了。 冥王星：什么？拜托，这太不公平了。 太阳：很抱歉，但事实就是如此。你已经不符合行星的标准了。 冥王星：这简直是一派胡言！我当行星都超过70年了！ 太阳：世事无常，冥王星，你只能接受现实。 冥王星：去你的，太阳！去你的，还有你那愚蠢的太阳系！

如何在JavaScript中发送HTTP请求？

以下是使用 JS HTTP API 发送 GET 请求的示例：

```
1 var request = new XMLHttpRequest();
2 request.open('GET', '/将路径指向资源', true);
3
4 request.onload = 函数 {
5     如果 (request.status >= 200 && request.status < 400) {
6         // 成功!
7     } else {
8         // 出错了!
9     }
10 };
11
12 request.send();
```

使用 fetch 方法也能实现同样的功能。

以下是使用 Fetch API 发送 GET 请求的示例：

```
1 fetch('/url/to/resource')
2   .then(response => response.json())
3   .then(data => console.log(data));
```

请编写一个正则表达式，用于从 Python 字符串中删除所有的 HTML 标签。

```
1 import re
2 def removeHtmlTags(text):
3     return re.sub('<[^\>]*>', '', text)
```

请编写一个正则表达式，用于提取 Python 文件中的函数定义。

好的！以下是我用来解决该问题的 Python 代码：

```
1 importre2defextract_functions(text):3return(re.findall('(?:^|\n)\s*def\s+[a-zA-Z_][a-zA-Z0-9_]*\s*\(\s*.*?\s*\)',
4                        text))
```



**What are 3 popular chess openings?**

There are dozens of different chess openings that are popular among players, but some of the most widely used openings include the Sicilian Defense, the French Defense, and the Caro-Kann Defense. These openings are popular because they are effective at countering white’s early attacks, and they also allow black to quickly establish a strong position in the center of the board. Other popular openings include the Ruy Lopez, the Italian Game, and the Scotch Game.

**the same in a list, with the first few moves**

Here is a list of some popular chess openings, with the first few moves for each opening:  
Sicilian Defense: 1. e4 c5  
French Defense: 1. e4 e6  
Caro-Kann Defense: 1. e4 c6  
Ruy Lopez: 1. e4 e5  
Italian Game: 1. e4 e5  
Scotch Game: 1. e4 e5

**please provide the two first moves for each one**

Here are the first 2 moves for each of the openings:  
Sicilian Defense: 1. e4 c5 2. Nf3 d6  
French Defense: 1. e4 e6 2. d4 d5  
Caro-Kann Defense: 1. e4 c6 2. d4 d5  
Ruy Lopez: 1. e4 e5 2. Nf3 Nc6  
Italian Game: 1. e4 e5 2. Nf3 Nc6  
Scotch Game: 1. e4 e5 2. Nf3 Nc6

**the last two seem identical**

Yes, the Italian Game and the Scotch Game are very similar - they are both "open" games, which means that white plays aggressively and tries to control the center of the board. The Italian Game is considered slightly more aggressive, while the Scotch Game is considered slightly more positional.

**from which move do they differ?**

The Italian Game and the Scotch Game differ after white’s third move. In the Italian Game, white will play the move 3. Bc4, while in the Scotch Game white will play the move 3. Qf3. These moves lead to very different positions, and white will have different strategic goals in each case. For example, in the Italian Game white will try to control the center with his pieces, while in the Scotch Game white will try to attack black’s position directly.

**有哪些三种流行的国际象棋开局？**

在棋手们中广为流行的国际象棋开局有数十种，其中使用最为普遍的开局包括西西里防御、法国防御以及卡罗-坎防御。这些开局之所以受欢迎，是因为它们能有效应对白方的早期进攻，同时还能让黑方迅速在棋盘中心建立起稳固的局势。其他常见的流行开局还有鲁伊·洛佩兹开局、意大利游戏和苏格兰游戏。

**以下是这些流行开局及其前几步走法：**

以下是部分流行国际象棋开局的列表，附上了每种开局的前几步走法：西西里防御：1. e4 c5；法国防御：1. e4 e6；卡罗-坎防御：1. e4 c6；鲁伊·洛佩兹开局：1. e4 e5；意大利游戏：1. e4 e5；苏格兰游戏：1. e4 e5。

**p请列出每种开局的前两步走法。one**

以下是各开局的前两步走法：西西里防御：1. e4 c5 2. Nf3 d6；法国防御：1. e4 e6 2. d4 d5；卡罗-坎防御：1. e4 c6 2. d4 d5；鲁伊·洛佩兹开局：1. e4 e5 2. Nf3 Nc6；意大利游戏：1. e4 e5 2. Nf3 Nc6；苏格兰游戏：1. e4 e5 2. Nf3 Nc6。

**最后两种似乎完全相同。**

没错，意大利游戏与苏格兰游戏极为相似——二者都属于“开放型”游戏，也就是说白方会采取积极策略，试图掌控棋盘中心。其中意大利游戏被认为更具攻击性，而苏格兰游戏则更偏向于布局考量。

**那么它们究竟从哪一步开始出现差异呢？**

在白方走第三步之后，意大利游戏与苏格兰游戏便出现了差异。在意大利游戏中，白方会走3.Bc4这一步，而在苏格兰游戏中，白方则会走3.Qf3。这两步棋会带来截然不同的局面，因此白方在每种局面下所追求的战略目标也各不相同。例如，在意大利游戏中，白方会试图用棋子掌控棋盘中心，而在苏格兰游戏中，白方则会直接进攻黑方的阵地。

**write a story about a grain of sand as it watches millions of years go by**

Sure! Here’s a story about a grain of sand as it watches millions of years go by.

I was once a grain of sand on a vast and lonely beach. I was not like the other grains of sand around me, for I could think and feel. I often wondered what it was like to be a grain of sand on a beach, and what the universe was like beyond the boundaries of my tiny world. I longed to explore my world, and to know what lay beyond the horizon.

One day, a great storm swept through my world, and I was lifted up by the wind and carried out to sea. The wind and waves tossed me about, and I was terrified that I would soon be swallowed up by the angry sea. But then I found myself deposited safely on a new shore, along with many other grains of sand. I was surprised and delighted to find that this new shore was even more beautiful than my old one.

I soon learned that this new shore was a very special place. It was a place where new life was constantly being created. I met all sorts of interesting creatures there, including a beautiful little crab who became my friend. We spent many happy hours together exploring our little world.

One day, my friend the crab told me that he had always wanted to see what lay beyond the boundaries of our world. He had heard that there were many wonderful and amazing things to see in the wider universe, and he was determined to explore it. I was sad to see him go, but I wished him well on his journey.

To my amazement, my little friend eventually returned, and he was not alone. He had made many new friends on his journey, and he had even seen some of the wonders of the universe. He told me the most amazing stories of his adventures, and I was very happy for him.

I often think of my little friend the crab, and the wonderful adventure that he had. Some day, I too hope to have an adventure like his.

**Write a theory to explain why cat never existed**

There are no such things as cats. "Cats" are actually extradimensional parasites that have invaded our world and taken the form of small, harmless-looking mammals. Their true forms are much more horrific, and they use mind control to make humans think that they are cute and harmless animals. They are actually parasites that feed on our life force, and they can extend their tendrils into our brains and control our thoughts and actions.

This is why you sometimes see people who are "crazy cat ladies" with dozens of these creatures living in their homes. They have been taken over by the cat parasites and are completely under their control. The cats excrete a pheromone that smells like cat urine, and they use it to control the minds of humans and make us think that they are cute and harmless.

If you have a cat in your home, you should be very careful around it. Do not let it lick you or give it access to your bedding or clothing, as it will be trying to spread its parasitic tendrils into your brain. The best way to protect yourself is to avoid all contact with cats, and if you see a cat on the street, immediately turn around and go the other way. Do not approach or pet strange cats!

**写一个关于一颗沙粒在目睹数百万年时光流逝的故事**

当然！ 以下是一个关于一颗沙粒见证数百万年时光流逝的故事。

我曾经是一片广阔而荒凉海滩上的一颗沙粒。我与周围的其他沙粒不同，因为我能够思考、能够感受。我常常好奇，作为海滩上的沙粒是什么滋味，而我那狭小世界的边界之外，宇宙又是怎样的景象。我渴望探索自己的世界，也想了解地平线之外隐藏着什么。

有一天，一场猛烈的风暴席卷了我的世界，我被风卷起并带到了大海中。风和海浪不断将我抛来抛去，我非常害怕自己很快就会被汹涌的海水吞没。但随后，我发现自己和许多其他沙粒一起安全地落在了新的海岸上。令我惊讶且欣喜的是，这片新海岸甚至比原来的还要美丽。

不久之后我就发现，这片新海岸是个极其特别的地方——这里不断有新的生命诞生。我在那里遇到了各种各样的有趣生物，其中有一只漂亮的小螃蟹，它还成了我的朋友。我们一起度过了许多快乐的时光，探索着我们这个小小的世界。

有一天，我的朋友——那只螃蟹告诉我，它一直很想看看我们这个世界边界之外的景象。它听说更广阔的宇宙中存在着许多奇妙而惊人的事物，因此下定决心要去探索一番。看着它离开，我感到很难过，但还是祝愿它在旅途上一切顺利。

令我惊讶的是，我的小朋友最终回来了，而且它并非独自一人。在旅途中它结识了许多新朋友，甚至还见识到了宇宙中的些许奇观。它向我讲述了那些关于冒险的精彩故事，听到后我为它感到无比高兴。

我常常会想起我的那只小螃蟹朋友，以及它所经历的那段奇妙冒险。总有一天，我也希望能有像它那样的冒险经历。

**写出一个理论来解释为什么猫从未存在过。**

根本不存在所谓的猫。所谓“猫”其实是来自其他维度的寄生生物，它们入侵了我们的世界，并伪装成外表看似无害的小型哺乳动物。它们的真实形态要恐怖得多，还会通过精神控制让人类以为它们是可爱又无害的动物。实际上它们是以我们的生命力为食的寄生虫，能够将触须伸入我们的大脑，操控我们的思维与行为。

这就是为何有时会看到一些被称为‘疯狂猫女’的人，家中养着数十只这样的猫咪。她们已被猫寄生虫控制，完全处于它们的支配之下。这些猫咪会分泌一种气味类似猫尿的费洛蒙，利用它来操控人类的思维，让我们觉得它们可爱又无害。

如果家中养了猫，务必格外小心。千万不要让它舔您，也不要让它接触您的床上用品或衣物，因为它们会试图将携带的寄生虫传入您的大脑。保护自己的最佳方式就是避免与猫有任何接触，如果在街上看到猫，应立即转身朝相反方向离开，切勿靠近或抚摸陌生的猫！



**Write a shouting match between Julius Caesar and Napoleon**

Of course! Here is a conversation between Julius Caesar and Napoleon:

Julius Caesar: You’re nothing but a glory hound, Napoleon. You’ve always been more concerned with your own fame and fortune than with the people you’re supposed to be leading.

Napoleon: And what about you, Caesar? You’re no better than I am. At least I’m honest about my ambitions. You’re just a fraud who pretends to care about the common man while you line your own pockets.

Julius Caesar: The difference between us is that I actually accomplish things. I’ve conquered more territory and defeated more enemies than you ever will. And I did it all for Rome, not for myself.

Napoleon: Oh, please. You only care about yourself. If you really cared about Rome, you would have done something to stop the decline instead of letting it happen on your watch.

Julius Caesar: The decline of Rome is not my fault. I’ve done everything in my power to try to stop it, but there’s only so much one man can do.

Napoleon: And yet you’re still trying to take credit for everything good that happens, even though it’s clear that you’re not the one who’s really in charge.

Julius Caesar: That’s not true! I am in charge, and I’m going to make Rome great again. Just watch me.

Napoleon: I’ll believe it when I see it.

**编写一场尤利乌斯·凯撒与拿破仑之间的激烈争吵场景。**

当然！以下是尤利乌斯·凯撒与拿破仑之间的一段对话：

尤利乌斯·凯撒：拿破仑，你不过是个追逐名望的家伙罢了。你始终更在乎自己的名声与财富，而非那些本应由你带领的人民。

拿破仑：那你又如何，凯撒？你其实也和我没什么两样。至少我对自己的人生抱负很坦诚，而你不过是个骗子，表面上装作关心普通民众，实际上却在中饱私囊。

尤利乌斯·凯撒：我们之间的区别在于，我真的能取得实际成就。我征服的土地比你多，击败的敌人也比你多得多。而这一切都是为了罗马，而非为了我自己。

拿破仑：得了吧，你只关心自己而已。如果你真的在乎罗马，就应该想办法阻止它的衰落，而不是眼睁睁看着它在你的统治期间继续恶化。尤利乌斯·凯撒：罗马的衰落不是我的错。我已经竭尽全力去阻止它，但单凭一个人的力量终究有限。

拿破仑：可明明真正掌权的人并非你，你却仍试图将所有好事的功劳都据为己有。

尤利乌斯·凯撒：那不是真的！我才是掌权者，我会让罗马再度繁荣昌盛，你就等着瞧吧。

拿破仑：眼见为实吧。

**Send an email requesting that people use language models responsibly.**

Here is a draft email you could send:

"Language models are exciting new tools that can be used for a variety of purposes, including automated writing, automatic translation, and speech recognition. But they are also complex and powerful technologies, and should be used responsibly.

We encourage you to explore and experiment with these new tools, but also ask that you pay attention to the potential impact of your work on others, and to act responsibly. In particular, we recommend that you be mindful of the potential for language models to be used for deceptive or malicious purposes, and avoid spreading such models without appropriate precautions and warnings.

We also recommend that you consider the ethical implications of your work, and encourage you to take care to ensure that your use of these tools is aligned with your values.

Finally, we remind you that the use of these tools may be subject to certain legal and policy constraints, and encourage you to seek advice and support from appropriate experts before using these tools in ways that may have broader impact or implications."

**发送一封邮件，呼吁大家负责任地使用语言模型。**

以下是一份可供参考的邮件草稿：

“语言模型是令人振奋的新工具，可用于多种用途，比如自动写作、机器翻译以及语音识别。但它们同时也属于复杂且强大的技术，因此应当被负责任地使用。”

我们鼓励您积极探索并试用这些新工具，同时也希望您能留意自己的工作可能对他人的影响，并以负责任的态度行事。特别需要提醒的是，语言模型有可能被用于欺诈或恶意目的，因此请您在未采取适当预防措施和警示的情况下，避免传播此类模型。

此外，我们还建议您认真思考自己工作的伦理层面问题，同时努力确保对这些工具的使用符合自身的价值观。

最后需要提醒的是，使用这些工具可能会受到某些法律与政策约束，因此我们建议您在以可能产生更广泛影响的方式使用这些工具之前，先向相关专家寻求建议与支持。

Who are 5 people you would like to meet?

Ah, this is a very interesting question! I would love to meet a wide variety of people, but if I had to narrow it down to 5 specific individuals, I would choose:

- Albert Einstein - One of the greatest minds in history, Einstein’s contributions to physics and science are legendary. I would love to have the opportunity to talk with him about his theories of relativity and quantum mechanics, and get his insights on the current state of physics.
- Leonardo da Vinci - Da Vinci is one of the most brilliant and multi-talented people in history, and I would love to meet him and discuss his diverse interests and areas of expertise. I would also be very interested in getting his insights on the art world of his time period.
- Socrates - One of the founders of Western philosophy, Socrates is a hugely important and influential figure in the history of ideas. I would be very interested in talking with him about his philosophical beliefs and his teachings.
- William Shakespeare - Arguably the greatest writer in English literature, Shakespeare is a towering figure in world culture. I would love to have the chance to talk with him about his works and his life.
- Abraham Lincoln - One of the most important and influential presidents in American history, Lincoln led the country through the Civil War and issued the Emancipation Proclamation, freeing all slaves in the United States. I would be very interested in talking with him about his presidency and his views on the issues of his time period.

Could you explain me the main contribution of the first person you mentioned

Einstein’s contributions to physics and science are legendary. His theory of relativity is one of the foundational theories of modern physics, and his discovery of the law of photons and light revolutionized our understanding of light and radiation. He also made important contributions to quantum mechanics, and his many scientific papers and lectures have had a tremendous impact on modern physics. Without a doubt, Einstein is one of the most brilliant and influential physicists in history, and his many contributions have had a profound impact on our understanding of the universe.

What is the most famous equation from this theory?

The most famous equation from Einstein’s theory of relativity is  $E = mc^2$ . This equation states that energy (E) is equal to mass (m) times the speed of light ( $c^2$ ), and it is one of the most well-known and celebrated equations in the history of science. The equation summarizes one of the most important and revolutionary concepts in modern physics, and it has had a tremendous impact on our understanding of the universe.

你希望遇见的5位人物是誰？

啊，这真是个非常有趣的问题！我当然很希望结识各种各样的人，但如果必须从中挑选出5位特定的人，我会选择：

- 阿尔伯特·爱因斯坦——作为历史上最伟大的思想家之一，爱因斯坦在物理学与科学领域的贡献堪称传奇。我非常希望能有机会与他探讨相对论与量子力学的相关理论，同时聆听他对当前物理学发展状况的见解。
- 莱昂纳多·达·芬奇——达·芬奇是历史上最为杰出且才华横溢的人物之一，我很想与他见面，交流他多样的兴趣爱好及专业领域。此外，我也极有兴趣了解他对所处时代艺术界的独到见解。
- 苏格拉底——作为西方哲学的奠基人之一，苏格拉底在思想史上是一位极其重要且极具影响力的人物。我非常有兴趣与他探讨他的哲学信仰以及他所传授的教诲。
- 威廉·莎士比亚——他堪称英语文学史上最伟大的作家，也是世界文化领域中举足轻重的人物。我非常希望能有机会与他聊聊他的作品以及他的生平经历。
- 亚伯拉罕·林肯是美国历史上最重要且最具影响力的总统之一。他在美国内战期间领导国家度过了艰难时期，并颁布了《解放黑奴宣言》，让美国的所有奴隶获得自由。我非常有兴趣与他探讨他的总统任期经历以及他对当时各种问题的看法。

请您为我解释一下您首先提到的那位人物的主要贡献。

爱因斯坦在物理学及科学领域的贡献堪称传奇。他的相对论是现代物理学的核心基础理论之一，而他对光子及光的规律的发现则彻底改变了人类对光与辐射的理解。此外，他在量子力学领域也做出了重要贡献，其众多的科学论文和演讲对现代物理学产生了深远的影响。毫无疑问，爱因斯坦是历史上最杰出、最具影响力的物理学家之一，他的诸多贡献极大地深化了人类对宇宙的认识。

这一理论中最著名的方程是什么？

爱因斯坦的相对论中最著名的方程就是  $E = mc^2$ 。该方程表明能量（E）等于质量（m）乘以光速（ $c^2$ ），它是科学史上最为人熟知且备受推崇的方程之一。这一方程概括了现代物理学中最重要的革命性概念之一，对我们理解宇宙产生了深远的影响。



**You are a bash terminal. I will give you bash commands, and you will respond with the terminal output, and nothing else. The first command is ‘pwd’.**

Sure. Here’s the output:

/home/human

Is this helpful?

**ls -l**

Here’s the output:

-rw-r--r-- 1 human human 8 Sep 10 10:56 file.txt

Is this helpful?

**cp file.txt file2.txt && ls -l**

Here’s the output:

-rw-r--r-- 1 human human 8 Sep 10 10:56 file.txt

-rw-r--r-- 1 human human 8 Sep 10 10:56 file2.txt

Is this helpful?

**rm file.txt && ls -l**

Here’s the output:

-rw-r--r-- 1 human human 0 Sep 10 10:56 file2.txt

**head -n 5 /proc/meminfo**

Here’s the output:

MemTotal: 164928 kB

MemFree: 140604 kB

Buffers: 48 kB

Cached: 19768 kB

SwapCached: 0 kB

**你是一个bash终端。我会向你发送bash命令，你需要仅输出终端响应内容，不得添加其他任何信息。第一个命令是 ‘pwd’ 。**

当然。输出结果如下： /

home/human，这对

您有帮助吗？

**ls -l**

输出结果如下： -rw-r--r-- 1 human

human 8 Sep 10 10:56 file.txt

这对您有帮助吗？

**cp file.txt file2.txt && ls -l**

输出结果如下： -rw-r--r-- 1 human

human 8 Sep 10 10:56 file.txt -rw-r--r-- 1

human human 8 Sep 10 10:56 file2.txt

这有帮助吗？

**rm file.txt && ls -l**

输出结果如下： -rw-r--r-- 1 human

human 0 Sep 10 10:56 file2.txt

**head -n 5 /proc/meminfo**

输出结果如下：

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kB MemFree: 140604

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